Dual Prompt Guidance for Cross-Scene Hyperspectral Anomaly Detection

Guosheng Li, Li Liu, Yingqian Wang, *Member, IEEE*, Chao Xiao, Caihuan Huang, and Kun Li

***Abstract*—Hyperspectral anomaly detection (HAD) aims to locate the pixels that have significant spectral deviations from their neighborhoods in an unsupervised manner. However, most existing deep learning-based HAD methods suffer from notable performance degradation when the models are transferred to new scenes and have to train the models from scratch. Therefore, their practical applications are restrained. To improve the generalization capability, we propose a dual prompt-guided training-free method for cross-scene HAD by introducing foundation models. Specifically, we formulate HAD as a pixel-wise segmentation task. A clustering-based band selection method is proposed to align the channels of hyperspectral images (HSIs) with the Segment Anything Model (SAM). The spectral bands closest to the cluster centers are selected as the input to SAM. Then, a dual prompt mechanism is proposed to distinguish anomaly from background. The scene-adaptive text prompts generate masks using the Grounding DINO model to indicate where the anomaly and background may be present. The point prompts provide the potential anomaly and background points refined by the masks, and guide the SAM to identify the anomalies. Experimental results demonstrate that our method achieves superior cross-scene detection performance to existing state-of-the-art HAD methods without additional training.**

***Index Terms*—Anomaly detection, dual prompt, foundation model, generalization, hyperspectral image (HSI).**

I. Introduction

THE unique fingerprint spectral characteristics of the hyperspectral image (HSI) [1], [2], [3], [4] make it well-suited to distinguish the subtle differences between various materials. Hyperspectral anomaly detection (HAD) aims to locate the pixels that have significant spectral deviations from their neighborhoods. It has been broadly applied in reality [5], [6], [7], [8], [9]. However, HAD is a challenging task due to the unknown spectral prior of the anomaly targets. Existing HAD methods usually use the statistic property [10], [11], [12], [13], low-rank prior of background [14], [15], [16], [17], and sparse property of anomaly [18], [19] to identify the anomaly targets.

Recently, the deep learning-based methods [7], [20], [21], [22], [23], [24], [25], [26] assume that anomalies have larger reconstruction errors than background, and achieve promising results. However, most of the existing deep learning-based methods highly depend on specific scenes. When the models are transferred to a new scene, the detection performance significantly degrades as the feature distributions of background and anomaly change. As shown in Fig. 1(a) and (c), the detection scenes in both HSIs are airports, and the anomaly targets are airplanes. However, due to the changes in the imaging environment, background structures, and spectral variations, the mean spectral curves of background and anomaly on these two scenes are notably different [see Fig. 1(b) and (d)]. These methods need to train the model from scratch in this case. Therefore, the generalization performance is limited and restricts their practical applications.

To tackle the aforementioned issue, we propose a dual prompt-guided cross-scene HAD method by introducing foundation models. Although the hyperspectral foundation models may address the generalization problem of HAD, this article focuses on the research under limited resource conditions. Since HAD is a pixel-wise binary classification task, it can also be viewed as a segmentation problem. Motivated by the powerful generalization capability of the Segment Anything Model (SAM) [27], we introduced it into cross-scene HAD. However, SAM is designed for natural images with three channels, while HSIs have hundreds of bands. How to align the HSI channels with SAM and how to reduce the domain shift between them need to be addressed.

Due to the band redundancy of HSIs, a clustering-based band selection strategy is proposed to align the channels of HSIs with SAM. The k-means++ method [28] is applied to the HSI bands to obtain the cluster centers. The spectral bands closest to the centers are selected as the input to SAM. Moreover, target detection in either an HSI or red-green-blue (RGB) image aims at distinguishing the target from the background via their feature differences. The target and background prompts can provide semantic information to guide SAM to perceive the feature differences. Therefore, a dual prompt mechanism (i.e., a text prompt and a point prompt) is proposed to bridge the gap between the HSI and natural image. Specifically, the vision-language generative model InstructBLIP [29] is used to describe the HSI and obtain an automatic prompt. Since InstructBLIP is also trained on natural images, the automatic prompt may neglect the characteristics of HAD. Considering the small size of the anomaly, a fixed prompt is used to enhance the capability of prompt representation. Both the automatic prompt and the fixed prompt are then fed into the Grounding DINO model [30] to generate boxes that indicate potential anomaly regions. By

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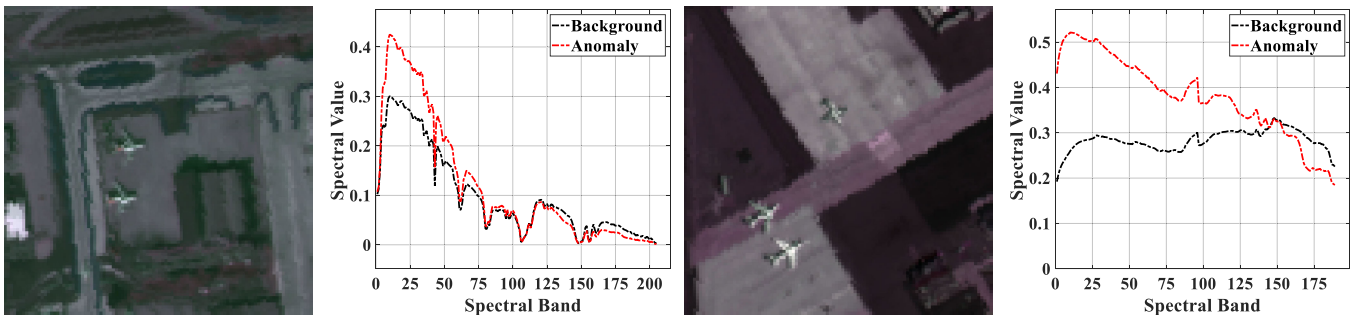

Fig. 1. Large spectral discrepancies of background and anomaly across scenes. (a) False-color image of Los Angeles dataset. (b) Mean spectral curves of background and anomaly on the Los Angeles dataset. (c) False-color image of San Diego-III dataset. (d) Mean spectral curves of background and anomaly on the San Diego-III dataset.

comparing the confidence of the generated boxes, a scene-specific text prompt is selected between the two prompts. The Reed-Xiaoli (RX) [10] algorithm generates initial anomalies and background. Afterward, the initial anomalies in the potential anomaly regions are refined as the anomaly prompt points, while the initial background points outside the anomaly regions are selected as background prompt points. It is worth noting that the proposed method is a training-free framework due to the frozen foundation models.

The major contributions of this article are summarized as follows.

1) We propose a dual prompt-guided training-free method for cross-scene HAD. By introducing the foundation models, our method can identify anomalies without training or fine-tuning the models when detection scenes change.
2) We propose text prompts and a point prompt strategy to distinguish anomalies from the background. The ensemble text prompts generate anomaly boxes to indicate the potential anomaly regions across various scenes. The point prompt is employed to obtain anomaly targets according to the text prompt.
3) We propose a clustering-based band selection strategy to align the channels of HSIs with foundation models due to the correlations among spectral bands. The bands closest to the cluster centers are selected as the input of the foundation models.

The remainder of this article is organized as follows. Section II briefly overviews related works. Section III elaborates on the details of our proposed method. Section IV presents experimental results and corresponding analyses. Section V concludes this article.

II. Related Works

A. Hyperspectral Anomaly Detection

HAD is a typical unsupervised task without prior spectral knowledge of the target [31], [32], [33], [34], [35]. Existing HAD methods mainly model the spectral deviations between each pixel and its neighborhoods to identify the anomalies. These methods are categorized into statistic-based methods, representation-based methods, matrix decomposition-based methods, and deep learning-based methods.

Statistic-based methods [10], [11], [12], [13] assume that the background spectra follow a specific statistic distribution RX [10] and its variants [11], [12], [13] serve as representative algorithms of this category, which take the Mahalanobis distance between the spectrum and the mean background spectrum as the anomaly score. However, complex backgrounds are difficult to characterize with a fixed data distribution in the original space. To address this issue, several methods perform HAD via domain transformation. For example, several researchers convert HSIs into the frequency space [36], kernel space [37], and topological space [38] for anomaly detection. Moreover, some methods [39], [40] first estimate the background, then identify pixels with large distances from the background as anomalies.

Representation-based methods [18], [19], [41] identify pixels that cannot be accurately linearly represented by background atoms as anomalies. Such methods can be further divided into sparse representation (SR)-based and collaborative representation (CR)-based methods. SR-based methods [18], [42] assume that background pixels can be sparsely and linearly represented by atoms in the background dictionary, while anomalous pixels yield large representation residuals. In contrast, CR-based methods [19], [41] hold that background pixels can be collaboratively represented by all pixels in their local neighborhoods, whereas anomalies cannot. Since only local features of HSIs are exploited, these methods exhibit unsatisfactory detection performance in scenes where anomaly targets are densely distributed.

The matrix decomposition-based methods [14], [15], [16], [17], [43], [44] assume that the background is low-rank, and anomaly is sparse in HSIs. The representation residuals are used as the anomaly scores. Existing methods [14], [15] improve the performance of anomaly detection by constructing a pure background dictionary. In addition, Cheng and Wang [43] introduce graph regularization to characterize the structural features of HSIs, enhancing the separability between background and anomalies. However, reshaping the HSI data into a matrix will lose the 3-D structural information of HSIs. To this end, tensor decomposition-based methods [45], [46],

[47] have been proposed to preserve the spatial neighborhood relationships and 3-D geometric structure of HSIs. Although these methods achieve promising detection performance, they rely on specific handcrafted features, which limits their applications.

The deep learning-based methods [48], [49], [50], [51], [52], [53], [54], [55], [56], [57], [58], [59], [60], [61], [62], [63], [64] reconstruct the HSIs and detect the anomalies by the reconstruction errors. Such methods employ autoencoders and their variants, assuming that the network can better reconstruct the dominant background. Furthermore, several methods adopt variational regularization [65], adversarial training [66], [67], and normalization flow [68] to constrain the latent representations, so as to avoid identity mapping and learn meaningful low-dimensional manifolds. Since anomalies can also be well reconstructed, several methods [20], [22], [69], [70] focus on anomaly suppression to improve the detection performance. Due to the poor interpretability of deep learning-based methods, some methods [71], [72], [73] also combine traditional methods with deep learning models to integrate prior knowledge and data-driven learning. However, most of the above methods only utilize spectral features and fail to fully exploit the inherent information of HSIs. Several methods further explore spatial features [74], [75], [76], graph structural features [22], [56], frequency-domain features [36], [77], [78], and endmember features [23], [24], [79] to enhance the feature learning. Nevertheless, most of them are performed on a single dataset that is captured from the same scene without dividing the training set and the test set. Therefore, the models are prone to overfitting across different scenes.

B. Cross-Scene Generalization in HAD

To enhance the generalization performance of the model in cross-scenario detection, Li et al. [80] utilized the Siamese network to learn the spectral similarity between positive and negative sample pairs on the reference dataset. When the model is transferred to the test dataset, the average spectral similarity between each pixel and its neighborhoods is used as the anomaly score. Rao et al. [21] used a clustering method to divide the test set into several clusters. The sample features in the same cluster are similar, while the sample features between different clusters are significantly different. They constructed positive and negative sample pairs from the clusters to fine-tune the Siamese network, and reduced the domain shift. However, this method neglects the impact of abnormal samples on clustering performance, which would affect the quality of sample pairs. Huyan et al. [81] learned the spectral correlation between each sample and its surroundings through few-shot learning. Li et al. [82] learned the spatial context relationship between the target and the background on pure background HSIs by adding random masks. Since it is difficult to exhaustively cover the mask positions, the generalization performance of the model is still limited. Jin et al. [83] used the reverse distillation method to train a background feature enhancement network on pure background images, which can more accurately estimate the background of new images.

C. Foundation Models

The Grounding DINO [30] model employs a multimodal decoupled architecture that comprises a visual encoder, a text encoder, a cross-modal feature enhancer, and a detection decoder. It integrates the Transformer attention and deformable attention mechanisms, and exhibits superior open-set and zero-shot object detection capabilities. By using text prompts, Grounding DINO can rapidly localize bounding boxes for arbitrary unseen targets, achieving cross-modal precise alignment. SAM [27] is a segmentation foundation model built on natural images with a Transformer attention mechanism. It consists of three decoupled modules: an image encoder, a prompt encoder, and a mask decoder. By simple prompts including point prompts, box prompts, and text prompts, SAM enables precise localization and extraction of masks for arbitrary targets in images. InstructBLIP [29] is a vision-language generative model that can translate visual content into textual answers following the user prompts. It consists of three parts, including a frozen vision encoder, an instruction-aware Query Transformer, and a frozen Large Language Model. Fu et al. [84] introduced the SAM to generate masks to represent the semantic information of HSIs, but they do not exploit the zero-shot and few-shot generalization performance of SAM. Transferring the foundation models to the HAD field for improving their generalization capability remains an open question. The hyperspectral foundation models, HyperFree [85] and HyperSIGMA [86], were developed recently. However, they may be incompatible with the downstream HAD task due to the loss of semantic information.

III. Proposed Methodology

A. Overview

In this article, a training-free framework guided by dual prompts is proposed for cross-scene HAD. The proposed method formulates the pixel-wise HAD as a segmentation task and introduces SAM as the backbone to obtain the anomaly scores. A dual prompt strategy is employed to incorporate semantic information for mitigating cross-scene feature shifts. The overall pipeline of the proposed method is illustrated in Fig. 2. The band selection module (BSM) first extracts three spectral bands from the HSI to construct a false-color image as the input of the foundation models. A text prompt generation module is proposed to produce an automatic prompt and a fixed prompt, and guide the Grounding DINO model to generate masks for locating potential anomaly regions. Subsequently, principal component analysis (PCA) is performed on the HSI, followed by the RX algorithm to obtain a coarse anomaly map. The point generation module (PGM) produces accurate positive and negative point prompts for SAM to obtain the anomaly map according to the anomaly masks. Finally, a postprocess approach is designed to further refine the anomaly map.

B. Band Selection Module

Due to the three-channel input requirement of foundation models, HSIs with tens to hundreds of spectral bands are not adapted to these models. The previous study [84] selects three

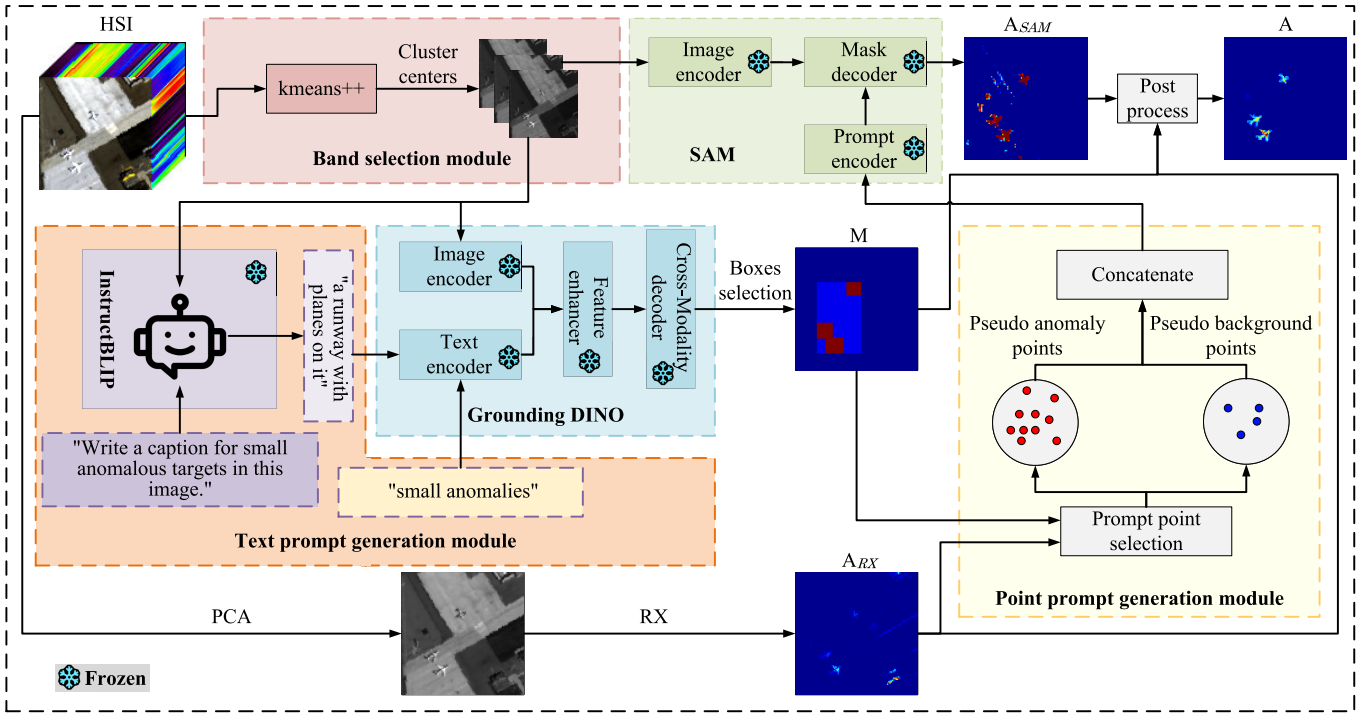

Fig. 2. Overall flowchart of the proposed network. The proposed method formulates the HAD as a pixel-wise segmentation task without network training. BSM is responsible for generating false-color images. The text prompt generation module produces an automatic text prompt and a fixed prompt to be fed into Grounding DINO, and generates mask maps to indicate the target region. The point prompt generation module produces point prompts for SAM to be aware of the anomaly and background. SAM serves as the segmentation backbone to output an anomaly map under the guidance of point prompts.

fixed bands from the HSI as the input of foundation models. However, fixed bands cannot reflect the spectral characteristics of different HSIs, leading to performance degradation across datasets. Although there exists band redundancy in the HSI, the spectral bands have high correlation. Therefore, a clustering-based band selection method is proposed to align the input channels of foundation models. Let $\mathbf{H} \in \mathbb{R}^{h\times w\times c}$ denote an HSI, where h, w, and c denote the height, width, and number of bands of the HSI, respectively. Specifically, due to its insensitivity to the initialization, the k-means++ clustering algorithm is applied to the HSI to segment it into three clusters along the spectral dimension. Subsequently, three bands closest to the cluster centers are selected from the obtained clusters to construct a false-color image $\mathbf{H}_{3c} \in \mathbb{R}^{h\times w\times 3}$. This band selection approach not only satisfies the input requirement of foundation models, but also preserves the spectral information of HSIs.

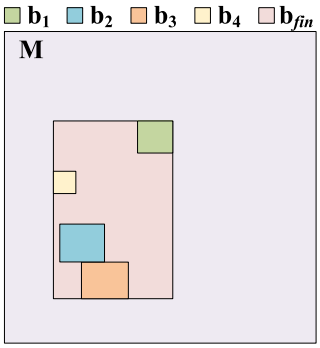

Fig. 3. Generation strategy of mask $\mathbf{M}$ on the San Diego-III dataset. The mask value of regions $\mathbf{b}_1, \mathbf{b}_2$, $\mathbf{b}_3$, and $\mathbf{b}_4$ generated by Grounding DINO is set to 1. The regions within the merging region $\mathbf{b}_{fin}$ while outside $\mathbf{b}_1, \mathbf{b}_2$, $\mathbf{b}_3$, and $\mathbf{b}_4$ are assigned with a mask value of α.

C. Text Prompt-Guided Mask Generalization

The text prompt serves as semantic guidance in the proposed method. To make the text prompt adaptive to various scenes, InstructBLIP is used to describe the detection scene via the text "Write a caption for small anomalous targets in this image." InstructBLIP can automatically generate a text caption $\mathcal{G}$ to describe the targets. However, due to the domain shift between natural images and HSIs, the automatic text prompt generated by InstructBLIP may fail to describe the HSIs in some complex scenes. Therefore, a fixed prompt $\mathcal{T}$ is designed to enhance the reliability of the aforementioned prompt. Since anomalies have a low probability of occurrence, their proportion is relatively low, and the size of targets is usually small in HSIs. Correspondingly, $\mathcal{T}$ is set as "small anomalies" according to the property of anomalies. Both the automatic text prompt and the fixed prompt are fed into the text encoder of Grounding DINO to generate text embeddings, which are fused with the image embeddings through a feature enhancer and cross-modality decoder. This cross-modal alignment enables the model to locate potential anomaly regions that match the semantic description of the prompt, thereby providing a precise anomaly prior for subsequent anomaly segmentation.

Specifically, the false-color image $\mathbf{H}_{3c}$, $\mathcal{G}$, and $\mathcal{T}$ are input into the Grounding DINO model to generate candidate boxes. Then, the model will select a more adaptive text prompt between $\mathcal{G}$ and $\mathcal{T}$ according to the confidence of the box generated by Grounding DINO. For prompts $\mathcal{G}$ and $\mathcal{T}$, the model generates a set of candidate boxes denoted as $B_{\mathcal{G}}$ and $B_{\mathcal{T}}$, respectively. Since the highest-confidence box typically covers the entire image, it is unsuitable for locating the anomaly regions. The boxes with the second-highest confidence (denoted as $s^*_{\mathcal{G}}$ and $s^*_{\mathcal{T}}$) are selected for comparison. The boxes corresponding to the higher confidence between $s^*_{\mathcal{G}}$ and $s^*_{\mathcal{T}}$ are adopted for subsequent mask generation

$$\mathcal{B}^* = \begin{cases} \mathcal{B}_{\mathcal{G}}, & s^*_{\mathcal{G}} \geq s^*_{\mathcal{T}} \\ \mathcal{B}_{\mathcal{T}}, & \text{otherwise} \end{cases} \tag{1}$$

where $\mathcal{B}^* = \{b_1, \ldots, b_n\}$ is the final selected box set, with each box $\mathbf{b}_i = (x_{i1}, y_{i1}, x_{i2}, y_{i2})$ defined by its top-left and bottom-right corners. The text prompt corresponding to $\mathcal{B}^*$ is regarded as more adaptive to the specific scene. Subsequently, box merging is performed on the generated bounding boxes, and the final unified bounding box $\mathbf{b}_{fin}$ is obtained by computing the outermost coordinates across all boxes in $\mathcal{B}^*$

$$\mathbf{b}_{fin} = \left(\min_{\mathbf{b}_i \in \mathcal{B}^*} x_{i1}, \min_{\mathbf{b}_i \in \mathcal{B}^*} y_{i1}, \max_{\mathbf{b}_i \in \mathcal{B}^*} x_{i2}, \max_{\mathbf{b}_i \in \mathcal{B}^*} y_{i2} \right). \tag{2}$$

Afterward, an anomaly mask $\mathbf{M} \in \mathbb{R}^{h \times w}$ is generated from the final box $\mathbf{b}_{fin}$ and the initial bounding box set $\mathcal{B}^*$. When multiple target boxes are present, the mask may produce overly strong constraints that tend to cause severe homogenization of background scores and degrade detection performance. Therefore, smoother mask weights are proposed to impose milder regional constraints and preserve the reasonable distribution characteristics of background scores. The value of each pixel (x, y) is defined as follows:

$$\mathbf{M}(x, y) = \begin{cases} 1, & (x, y) \in \mathbf{b}_i \text{ for any } \mathbf{b}_i \in \mathcal{B}^* \\ \alpha, & (x, y) \in \mathbf{b}_{fin} \text{ and } (x, y) \notin \bigcup_{i=1}^{n} \mathbf{b}_i \\ 0, & \text{otherwise} \end{cases} \tag{3}$$

where α is a hyperparameter that controls the weight of regions outside the boxes generated by Grounding DINO. The mask is used to highlight the anomaly and suppress the background simultaneously. The mask generation strategy on the San Diego-III dataset is illustrated in Fig. 3. Note that the Grounding DINO model is frozen.

D. Point Prompt Generation Module

HAD is a pixel-wise binary classification task of background and targets. It can be regarded as a segmentation task. SAM is introduced to segment the HSI due to its powerful generalization capability. However, there exists a significant domain gap between natural images and HSIs. Directly applying SAM to anomaly segmentation on false-color images of hyperspectral data cannot achieve satisfactory performance. To reduce the domain shift, a point prompt strategy is proposed to enable the SAM to perceive anomalies and the background. Specifically, the PCA is performed on $\mathbf{H}$ to suppress noise interference, yielding $\mathbf{H}_{pca} \in \mathbb{R}^{h \times w \times r}$ after dimension reduction. Then, the RX algorithm is applied to $\mathbf{H}_{pca}$ to generate an anomaly detection map $\mathbf{A}_{RX}$. A straightforward strategy is to select point prompts from $\mathbf{A}_{RX}$. However, anomaly detection scores achieved by RX are sensitive to the complex background distributions, which will result in inaccurate point prompts.

To improve the reliability of point prompts, a prompt point selection strategy is proposed to generate pseudo anomaly and pseudo background prompt points. Specifically, the top m points with the highest anomaly scores within the nonzero regions of $\mathbf{M}$ are selected as positive prompt points. Their coordinates are obtained, and corresponding labels are set to 1. In contrast, the n points with the lowest anomaly responses within the zero-valued regions of $\mathbf{M}$ are selected as background prompt points. The labels of these background prompt points are set to 0. Finally, the obtained anomaly and background prompt points are concatenated as the point prompts. The detailed formulation is given as follows:

$$\begin{cases} \mathcal{P}_a = \underset{\substack{|\mathcal{P}_a|=m \\ \mathbf{M}(x,y)\neq 0}}{\arg\max} \mathbf{A}(x, y), & l_a = 1 \\ \mathcal{P}_b = \underset{\substack{|\mathcal{P}_b|=n \\ \mathbf{M}(x,y)=0}}{\arg\min} \mathbf{A}(x, y), & l_b = 0 \\ \mathcal{P} = Concat(\mathcal{P}_a, \mathcal{P}_b) \end{cases} \tag{4}$$

where $\mathcal{P}_a$ and $\mathcal{P}_b$ denote the sets of anomaly and background prompt points selected from the nonzero and zero regions of $\mathbf{M}$, respectively. $l_a = 1$ and $l_b = 0$ represent their corresponding targets and background pseudolabels. $\mathcal{P}$ is the concatenated set of all prompt points. $|\cdot|$ denotes the number of elements contained in the set.

E. Dual Prompt-Guided Segmentation

The proposed method leverages precise target point prompts to guide SAM in segmenting the anomalies, and uses the logit output of SAM as anomaly scores. The higher scores indicate a higher probability of being anomalies, while lower scores correspond to the background. Specifically, the $\mathbf{H}_{3c}$ and $\mathcal{P}$ are fed into SAM to obtain the final segmentation mask $\mathbf{A}_{SAM} \in \mathbb{R}^{h \times w}$. A postprocessing strategy is designed on $\mathbf{A}_{SAM}$ to further highlight the targets and suppress the background. $\mathbf{A}_{SAM}$, $\mathbf{M}$, and $\mathbf{A}_{RX}$ are all incorporated into the computation. The final anomaly score map $\mathbf{A}$ is then calculated as follows:

$$\mathbf{A} = \mathbf{A}_{SAM} \odot \mathbf{A}_{RX} \odot (\mathbf{M} + \alpha(1 - \mathbf{M})). \tag{5}$$

The hyperparameter α ensures that anomalies can still be reliably detected by the model when the bounding box is inaccurate. By leveraging the guidance of the dual prompts, the proposed method can automatically perceive the anomaly and background across scenes. Note that the proposed method is a training-free framework because the SAM and Grounding DINO models are frozen.

Dual prompts provide a complementary mechanism to achieve more accurate anomaly detection. Specifically, text prompts uncover the potential anomalies and reduce the search space, providing prior regional knowledge for subsequent localization. Point prompts further refine target localization according to such region prior. In the absence of text prompts, point prompts may drift away from true anomalous regions to

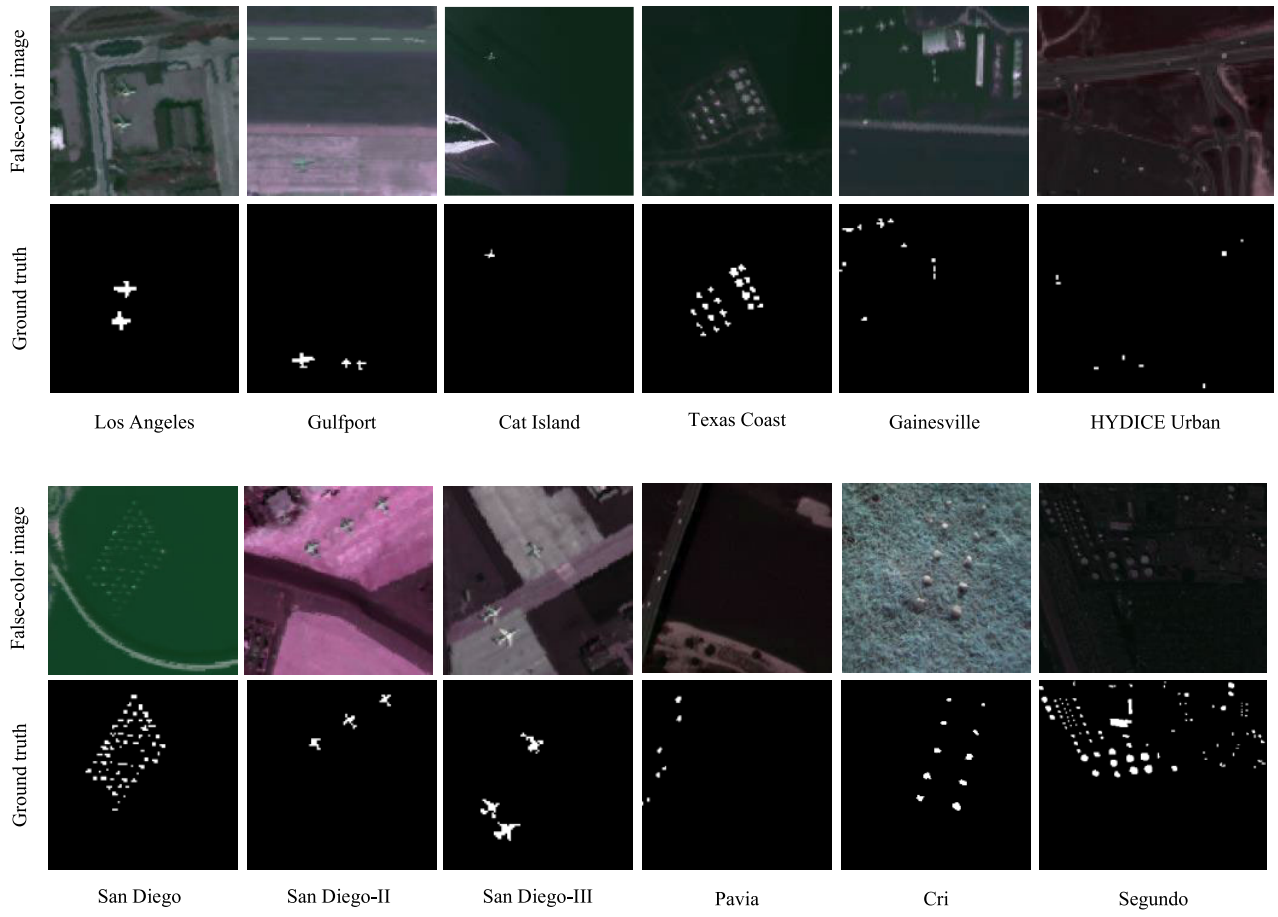

Fig. 4. False-color and Ground-truth images of 12 experimental datasets.

TABLE I
DETAILS OF THE 12 EXPERIMENTAL DATASETS

Dataset	Sensor	Spatial resolution	Wavelength range	Image size	Anomaly targets	Anomaly size	Anomaly proportion
Los Angeles	AVIRIS	7.1m/pixel	400-860 nm	$100 \times 100 \times 205$	Airplanes	87	0.87%
Gulfport	AVIRIS	3.4 m/pixel	400-2500 nm	$100 \times 100 \times 191$	Airplanes	60	0.60%
Cat Island	AVIRIS	17.2 m/pixel	400-2500 nm	$150 \times 150 \times 188$	Ships	19	0.08%
Texas Coast	AVIRIS	17.2 m/pixel	400-2500 nm	$100 \times 100 \times 207$	Buildings	155	1.55%
Gainesville	AVIRIS	3.5 m/pixel	400-2500 nm	$100 \times 100 \times 191$	Ships	52	0.52%
HYDICE Urban	HYDICE	1.56 m/pixel	400-2500 nm	$80 \times 100 \times 162$	Vehicles	17	0.21%
San Diego	AVIRIS	7.5 m/pixel	400-2500 nm	$100 \times 100 \times 193$	Ships	202	2.02%
San Diego-II	AVIRIS	3.5 m/pixel	370-2510 nm	$100 \times 100 \times 224$	Airplanes	56	0.56%
San Diego-III	AVIRIS	3.5 m/pixel	370-2510 nm	$100 \times 100 \times 191$	Airplanes	134	1.34%
Pavia	ROSIS-03	1.3 m/pixel	430-860 nm	$150 \times 150 \times 102$	Vehicles	68	0.30%
Cri	Nuance Cri	unknown	650-1100 nm	$400 \times 400 \times 46$	Rocks	1254	0.78%
Segundo	AVIRIS	7.1 m/pixel	366-2496 nm	$250 \times 300 \times 224$	Oil storage tank	2048	2.73%

complex-background pixels. Without point prompts, the proposed model cannot refine the potential anomaly localizations by using only coarse anomaly regions, preventing SAM from accurate segmentation. In contrast, the dual prompt mechanism improves anomaly detection performance.

IV. EXPERIMENTS

We perform experiments on 12 public datasets across different scenes to validate the effectiveness of our method, including Los Angeles,[1] Gulfport,[1] Cat Island,[1] Texas Coast,[1] Gainesville,[1] Hyperspectral Digital Imagery Collection Experiment (HYDICE) Urban [23], San Diego,[1] San Diego-II [23], San Diego-III [58], Pavia,[1] Cri [31], and Segundo [22]. The false-color images and ground truth of these datasets are shown in Fig. 4. More details about these datasets are shown in Table I.

A. Experiment Settings and Evaluation Metrics

We compare the proposed method with existing state-of-the-art HAD algorithms, including statistic-based method RX [10], representation-based method learnable background endmember with subspace representation (LEBSR) [15], deep learning-based methods AutoAD [20], BockNet [58], gated transformer network for HAD (GTHAD) [59], MSNet [60], endmember-guided spectral-spatial feature diffusion (EGSSFD) [24], opti-

[1]http://xudongkang.weebly.com/

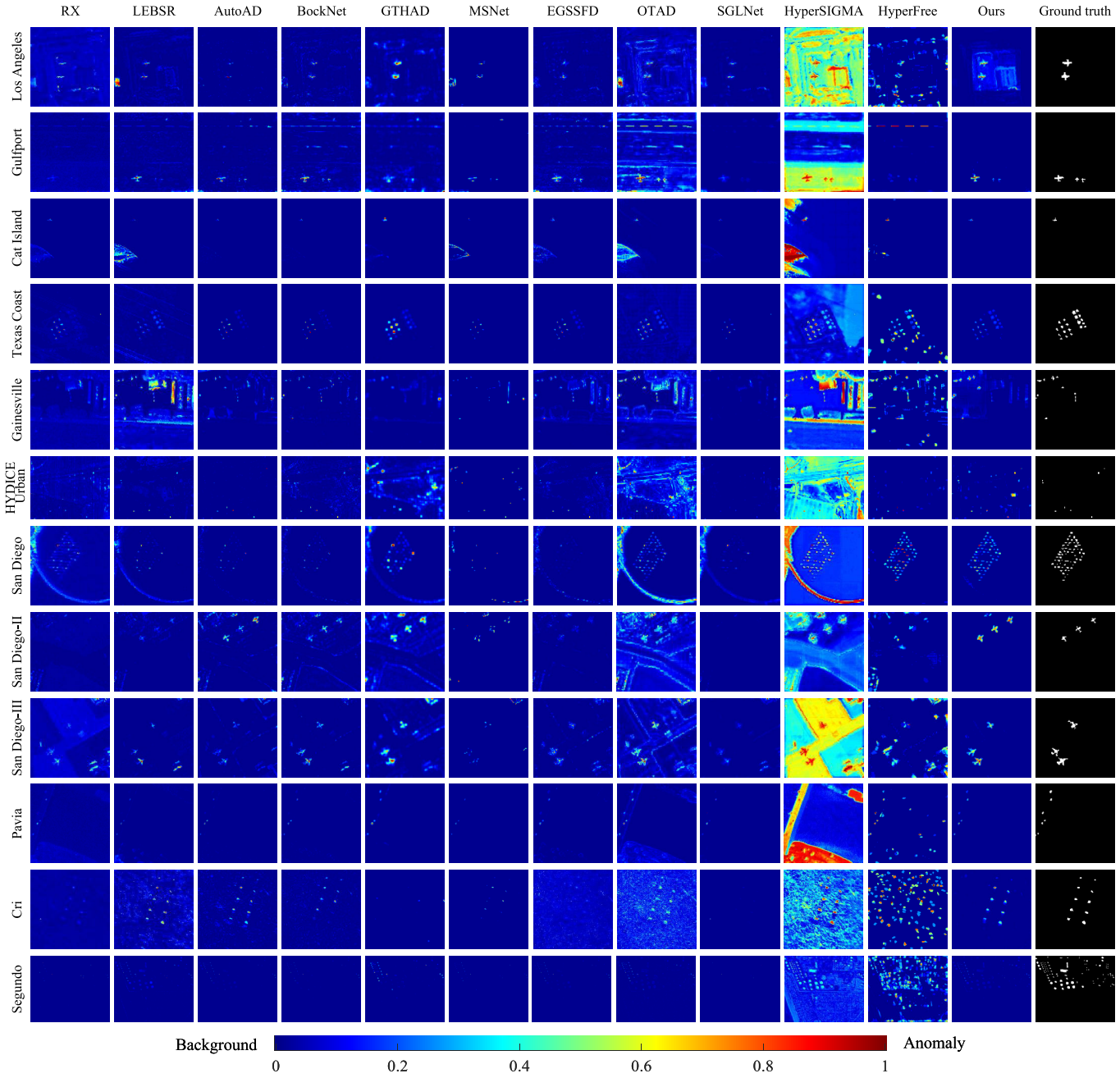

Fig. 5. Anomaly detection maps of different methods on experimental datasets.

mal transport-guided Transformer anomaly detector (OTAD) [61], SGLNet [56], and foundation model-based methods HyperFree [85] and HyperSIGMA [86]. All compared methods are implemented using their official codes. Since the foundation models used in our method are frozen, our method does not require training. As general-purpose foundation models, HyperFree and HyperSIGMA require fine-tuning for HAD to ensure a fair comparison. Both of them leverage RX-derived detection maps as guidance, but different fine-tuning strategies are performed on them due to different model structures. HyperFree updates the mask decoder while freezing all other modules, whereas HyperSIGMA performs full fine-tuning following its official configuration.

The anomaly detection map, receiver operating characteristic curve (2D-ROC), box plot [87], 3D-ROC curve [70], area under ROC curve area under the curve (AUC)(P_d, P_f), AUC(P_f, τ), and AUC(P_d, τ) are used for performance evaluation. Specifically, the anomaly detection map reflects the degree of target response and background suppression. 2D-ROC reflects a balance between true positive rate (TPR) and false alarm rate (FAR). Correspondingly, AUC(P_d, P_f) quantitatively measures the overall performance of the model to distinguish background from anomalies. A higher AUC(P_d, P_f) score means better performance. AUC(P_f, τ) intuitively reflects how the FAR changes with the decision threshold. The smaller AUC(P_f, τ) score reflects better background

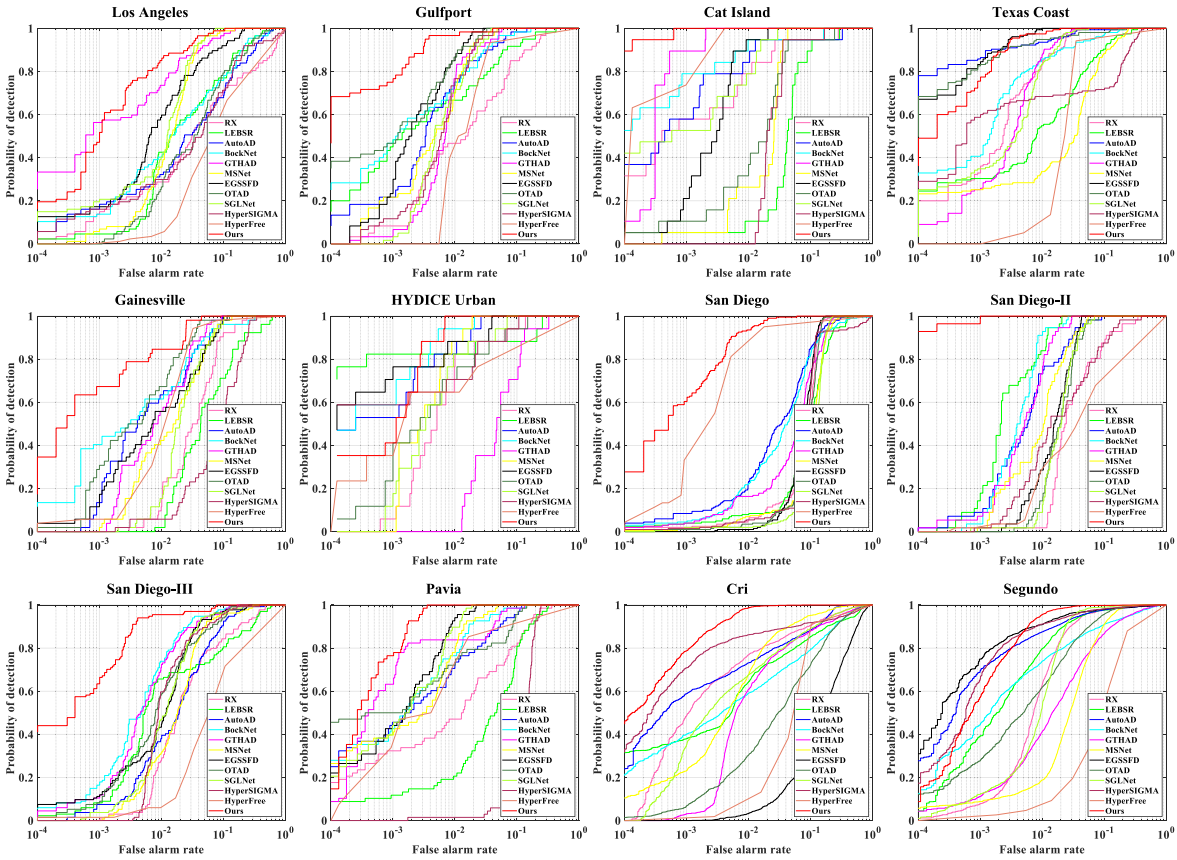

Fig. 6. 2-D ROC curves of (P_d, P_f) of different methods on experimental datasets.

suppression performance. AUC(P_d, τ) characterizes how the detection rate changes with τ. A larger AUC(P_d, τ) indicates that anomalies are more prominently highlighted by the model.

B. Detection Performance

1) Qualitative Performance: The anomaly maps are shown in Fig. 5. The compared methods can perform well on either background suppression or target enhancement. Traditional methods, including RX and LEBSR, suffer from significant performance degradation, especially on the Gainesville, San Diego, and San Diego-III datasets with complex backgrounds. Deep learning-based methods exhibit better robustness when dealing with complex scenes. However, they fail to effectively distinguish anomalies from the background on datasets such as Gulfport, San Diego, and San Diego-III, resulting in undesirably high anomaly scores in background regions of the detection maps. The foundation model HyperSIGMA achieves remarkable target response on all datasets. Nevertheless, it incorrectly classifies certain background regions as anomalies on the Cat Island, Gainesville, and San Diego datasets, and thus degrades the detection performance. HyperFree achieves satisfactory detection performance on the San Diego dataset without additional manual prompts. However, the automatically generated point prompts may fall into the background regions on the Cat Island, Texas Coast, and Gainesville datasets. This causes the model to misclassify these background regions as anomalies, and thus degrades the detection performance. In contrast, the proposed method simultaneously achieves accurate anomaly responses and effective background suppression without training or parameter tuning on all experimental datasets. Particularly, the anomaly score maps on the Cat Island, San Diego-II, San Diego-III, Pavia, and Cri datasets are highly consistent with the ground truth.

The 2-D ROC curves of (P_d, P_f), (P_f, τ), and (P_d, τ) are presented in Figs. 6–8, respectively. For the 2-D ROC curves of (P_d, P_f), the curves of the proposed method lie closest to the top-left corner on most datasets, indicating superior detection performance. Although the curve of our method intersects with that of GTHAD on the Los Angeles dataset, it still reaches the maximum detection rate first. Meanwhile, our method achieves competitive results with EGSSFD on the Texas Coast dataset. In terms of the ROC curves of (P_f, τ), the proposed method obtains the curves closest to the bottom-left corner on the Gulfport, Cat Island, San Diego, San Diego-III, and Pavia datasets, yielding the optimal background suppression performance. For the (P_d, τ) curve on the San Diego-II dataset, our curve is closest to the top-right corner, which achieves the best target enhancement effect. Furthermore, competitive performance is also maintained on the remaining experimental datasets. The 3-D ROC curves are illustrated in Fig. 9. Our method achieves a high detection rate with a low FAR on most datasets. In addition, it maintains stable and superior performance in background suppression and anomaly target

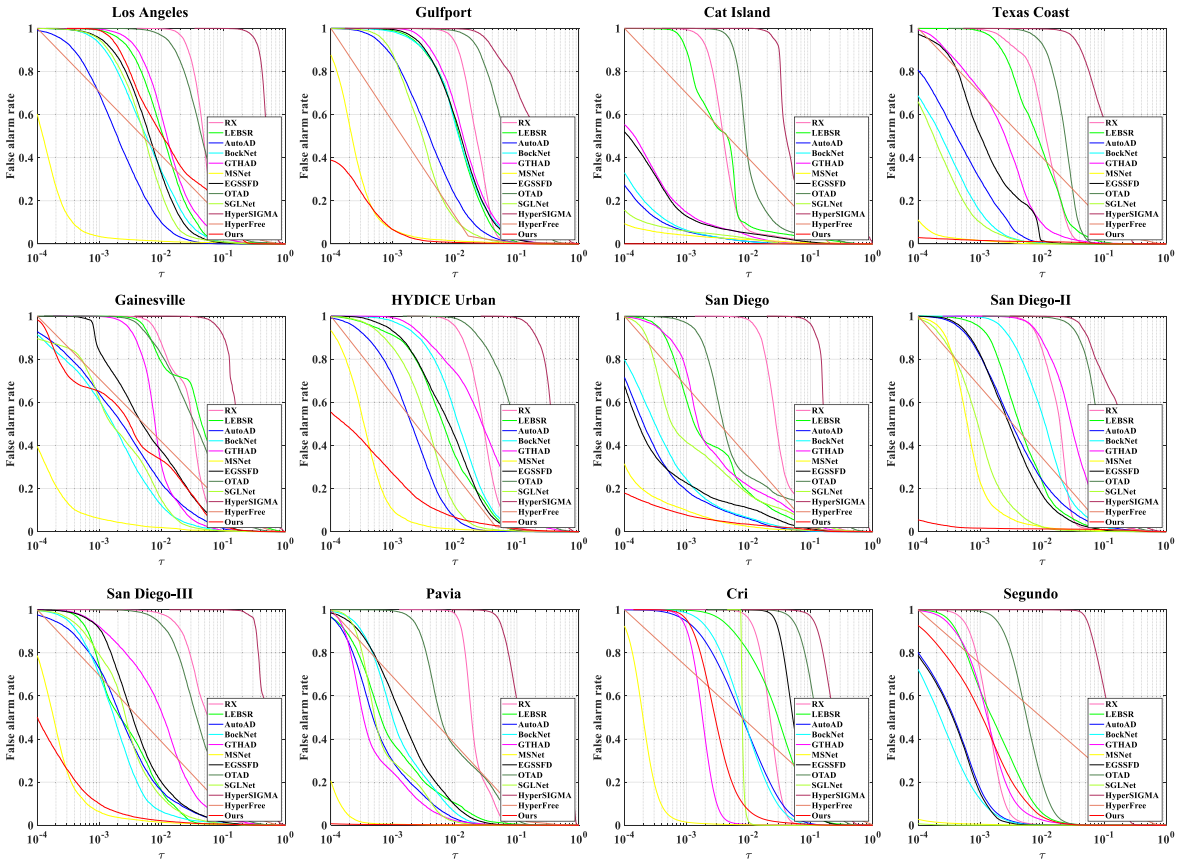

Fig. 7. 2-D ROC curves of (P_f, τ) of different methods on experimental datasets.

enhancement under diverse scenarios. This demonstrates the effectiveness and robustness of the proposed method.

The box plots of all comparison algorithms are shown in Fig. 10. In this article, we set the box range of the box plot to 25% and 75%, and the upper and lower whiskers to 95% and 5%, respectively. The proposed method achieves excellent background suppression by leveraging the dual prompts. On all datasets except Los Angeles and Gainesville, the background boxes of our method converge almost to a straight line, demonstrating the optimal background suppression performance. On the Los Angeles and Gainesville datasets, although the background boxes of our method exhibit a moderate degree of dispersion, they remain within an acceptable range. Furthermore, the background and anomaly boxes obtained by our method are completely separable on all datasets, verifying its excellent background-anomaly separability.

2) Quantitative Performance: The AUC scores are reported in Tables II and III. The optimal AUC scores are bold, and the suboptimal results are underlined. We draw the following observations.

1) Although the compared models are trained on each dataset, our method achieves optimal AUC(P_d, P_f) scores on most datasets without training, as shown in Table II. Compared with the suboptimal result, our method increases the AUC(P_d, P_f) values by 0.59%, 0.30%, 0.04%, 0.64%, 0.05%, 2.60%, 0.45%, 0.84%, 0.26%, 1.97%, and 0.58% on eleven datasets, respectively. Meanwhile, its performance on the Texas Coast dataset is comparable to that of the EGSSFD method. Therefore, our method has more robust generalization capability.
2) To validate the detection of the HAD methods across scenes, we train the deep learning-based methods on a dataset and test them on the remaining datasets. The band alignment strategy is used in [88]. As shown in Table III, the detection performance of the compared methods significantly degrades. RX, LEBSR, and ours do not require additional training. Their detection results are obtained by inference on the test datasets.
3) Due to the lack of semantic information, both the HyperFree and HyperSIGMA models are not adapted well to the downstream HAD task. The AUC(P_d, P_f) scores achieved by the two general foundation models are much lower than those achieved by our method, which validates the effectiveness of the dual prompts.
4) Since HyperSIGMA adopts a block-based strategy and guides the fine-tuning process with the cross-entropy loss, it can better highlight the targets and achieve higher AUC(P_d, τ) values on most datasets. However, it cannot balance false alarm suppression and target detection well.

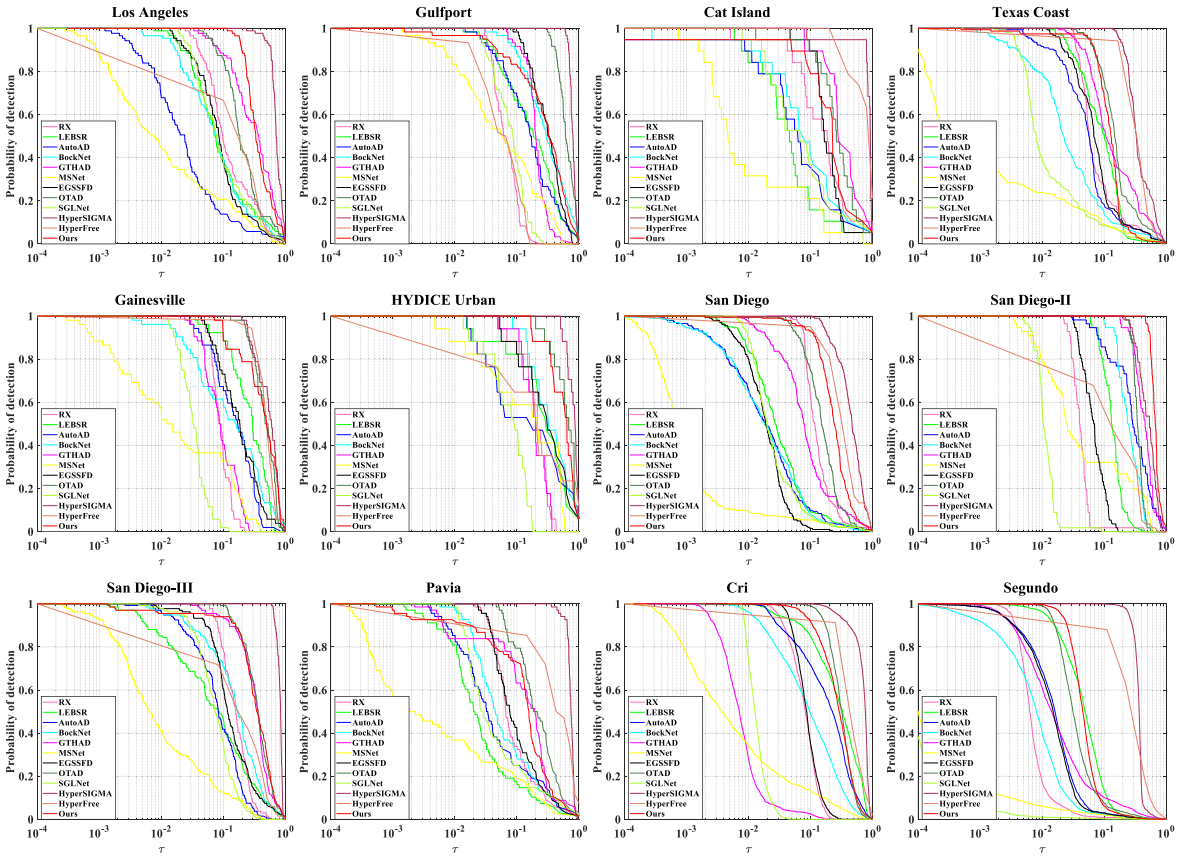

Fig. 8. 2-D ROC curves of (P_d, τ) of different methods on experimental datasets.

5) Our method employs masks generated from text prompts to suppress the background, thereby achieving satisfactory AUC(P_f, τ) scores on most datasets. In particular, the AUC(P_f, τ) scores achieved by our method approach 0 on the Gulfport, Cat Island, Texas Coast, and Pavia datasets. MSNet introduces a new separate training strategy and also yields satisfactory background suppression performance.

C. Model Analyses

1) Analysis of RX in the Point Prompt: The detection results of RX are used to obtain the point prompt. However, RX cannot achieve satisfactory detection results when the background has a complex distribution. We conduct experiments to analyze the robustness of our model when RX fails on the Gulfport, San Diego, San Diego-III, and Pavia datasets. As shown in Fig. 11, although RX fails to locate the anomalies on these datasets, the mask generated by our method imposes a spatial constraint on anomaly location, and thus yields satisfactory detection results.

2) Analysis of Scale Description in the Fixed Prompt: We use varying scale descriptions in the fixed text prompt (i.e., "small anomalies") on the San Diego-II and San Diego-III datasets to analyze its impact on detection performance. The experimental results are presented in Fig. 12. The results indicate that the scale description affects detection performance. The proposed method suffers from a notable degradation of performance when using the prompt "large anomalies." This is because the prompt is inconsistent with the small size of hyperspectral anomalies. Since "small anomalies" matches the inherent sparsity of anomalies in HSIs and achieves favorable detection performance, we select it as the fixed prompt.

3) Analysis of SAM: We remove the SAM in our method to quantitatively evaluate the benefit brought by it. The anomaly score map is obtained by the element-wise multiplication of $\mathbf{A}_{RX}$ and $\mathbf{M}$. The corresponding experimental results are shown in Table IV. It can be observed that SAM significantly improves the detection performance on all datasets. This is because the point prompt in SAM can aid in identifying the anomalies.

4) Analysis of Box Prompt in SAM: The bounding boxes produced by Grounding DINO can also be fed into SAM as a box prompt. We design additional comparative experiments to investigate the impact of such a box prompt on detection accuracy. The corresponding experimental results are shown in Table V. It can be observed that incorporating box prompt into SAM does not yield performance improvement, because the background regions inside the prompt boxes disrupt the accurate anomaly segmentation.

5) Analysis of Model Parameter Size: To evaluate the impact of model parameter size on detection performance, we experiment with foundation models of different parameter sizes. Specifically, we utilize three variants for SAM, including the base version (94 M), large version (312 M), and huge

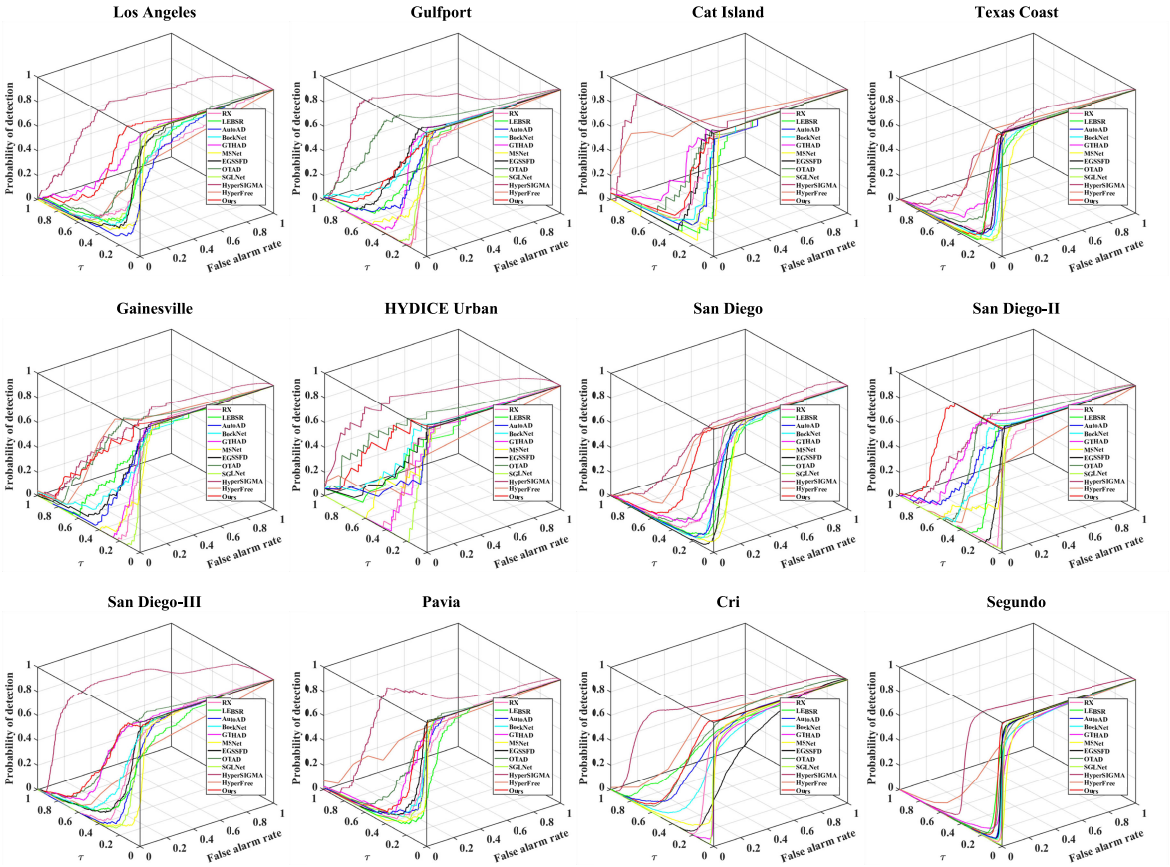

Fig. 9. 3D-ROC curves of different methods on experimental datasets.

version (641 M). For Grounding DINO, we select the tiny (173 M) and base (233 M) versions. For InstructBLIP, we only adopt the FlanT5-XL variant (3B) for inference, because larger parameter versions will lead to prohibitive inference time costs. The experimental results are shown in Table VI. The combination of SAM-Base and Grounding DINO-Tiny achieves the optimal detection performance. Overall, increasing model parameter size tends to degrade the detection performance.

6) Analysis of the Postprocessing Strategy: To validate the effectiveness of the postprocessing strategy in (5), we design four variants: 1) without $\mathbf{A}_{SAM}$; 2) without $\mathbf{A}_{RX}$; 3) without $\mathbf{M}$; and 4) without both $\mathbf{A}_{RX}$ and $\mathbf{M}$. The experimental results are presented in Table VII. Our method achieves superior detection performance to the postprocessing variants on most datasets. Removing either the $\mathbf{A}_{\mathrm{SAM}}$ term or the $\mathbf{A}_{\mathrm{RX}}$ term fails to enhance the anomaly scores. Removing the $\mathbf{M}$ loses the spatial location constraint on the anomaly targets.

7) Analysis of Blurred-Boundary Detection Performance: To validate the robustness of our method against scenes with indistinct boundaries between anomalies and backgrounds, we conduct additional experiments on the Bay Champagne [31] and the synthetic HyMap [24] datasets with blurred target boundaries. The Bay Champagne dataset is captured by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor and has an image size of $100 \times 100 \times 188$ and spans the spectral range from 400 to 2500 nm. The spatial resolution of this dataset is 4.4 m/pixel. The anomalous target in the scene is a ship, which corresponds to 17 anomaly pixels. The synthetic HyMap dataset is collected by the HyMap sensor and has an image size of $100 \times 150 \times 126$. The spatial resolution of this dataset is 3 m/pixel. It is also worth noting that the anomalies consist of 12 artificially synthesized low-abundance subpixel targets arranged in 3 rows and 4 columns on the HyMap dataset. The abundance of targets in each column is set to 0.05, 0.3, 0.2, and 0.1 from left to right, respectively. The target sizes in HyMap are 1×1, 2×2, and 3×3 from top to bottom, respectively. The boundaries between anomalous targets and the background are blurred. The false-color image and ground truth of the two datasets are shown in Fig. 13. The corresponding quantitative results are shown in Table VIII. It can be observed that our method achieves the optimal detection performance on both datasets. The detection results in Fig. 13 demonstrate that our algorithm can accurately localize anomalies while effectively suppressing background interference. The experimental results further validate that our algorithm obtains favorable detection performance on scenes with blurry target boundaries.

D. Ablation Study

We conduct an ablation study to investigate the effectiveness of the proposed band selection and prompt strategies. For simplicity, the automatic text prompt, fixed text prompt, and point prompt are denoted as ATP, FTP, and PP, respectively.

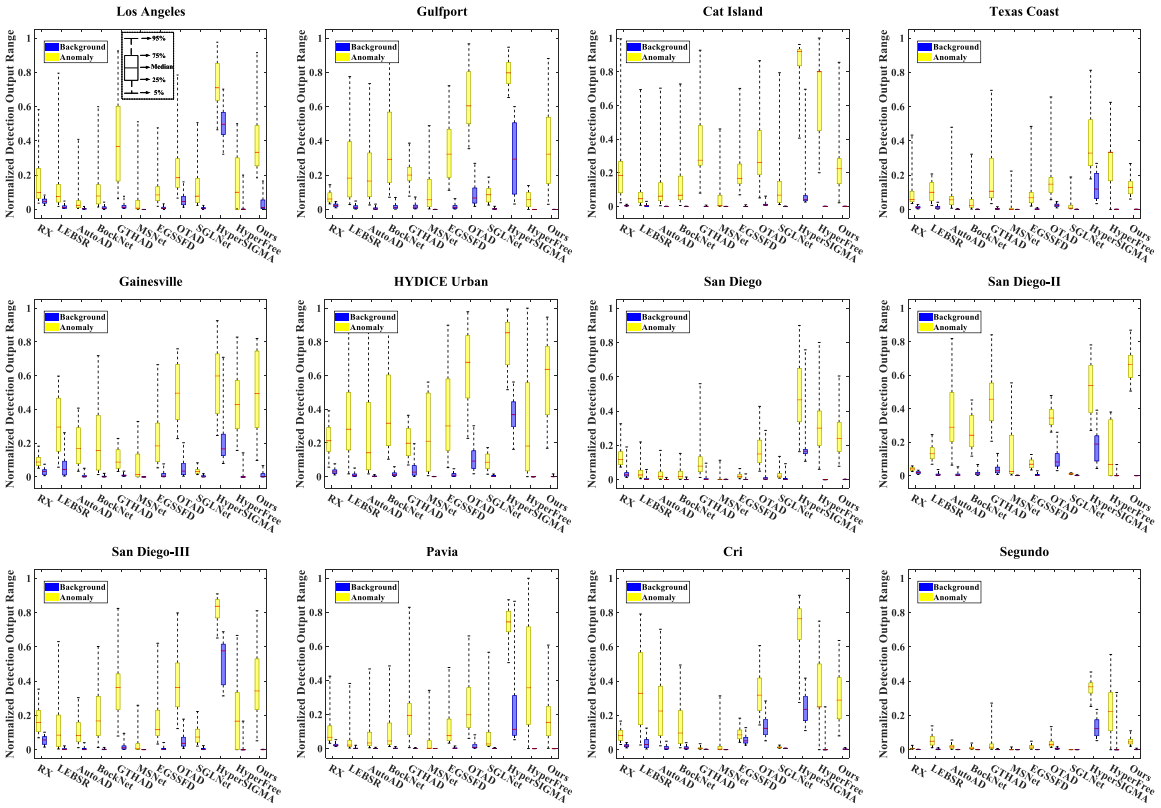

Fig. 10. Box plots of different methods on experimental datasets.

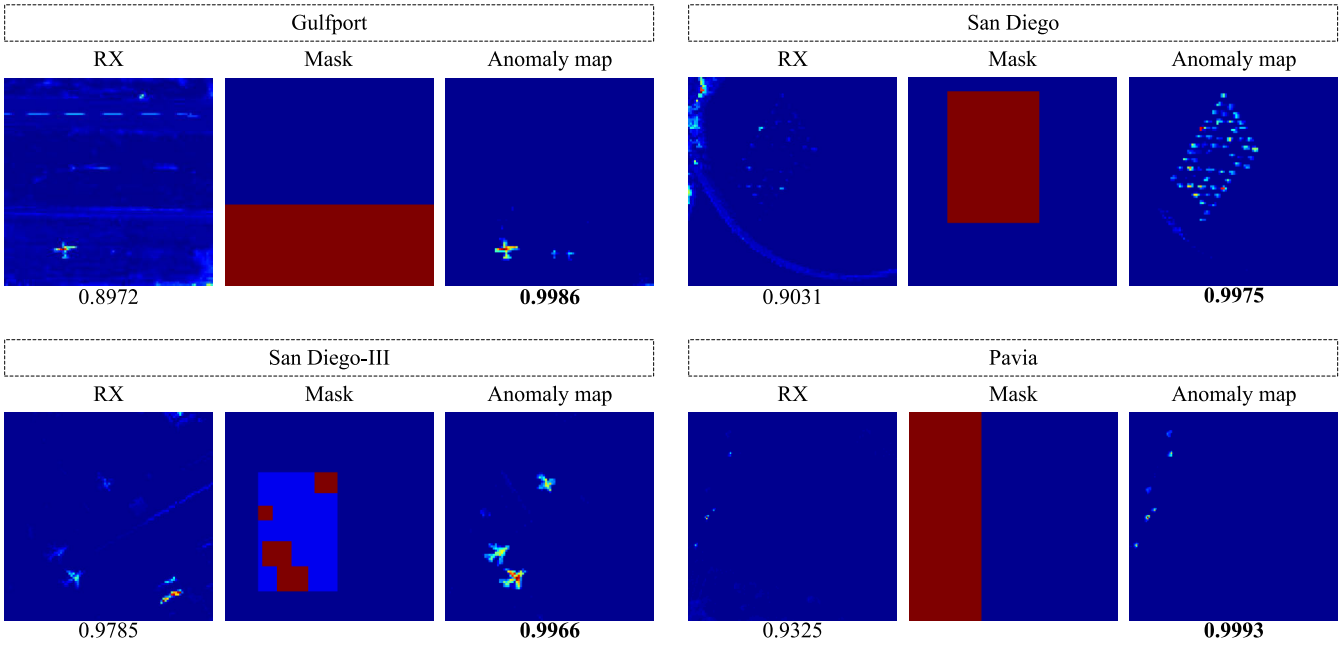

Fig. 11. Spatial constraint of the mask when RX fails to locate the anomalies.

1) Effect of the Band Selection for SAM: To investigate the effectiveness of the band selection strategy, we perform PCA and random selection to select three channels of HSIs as the input of SAM. For the random selection-based band selection method, we use three different random seeds (1, 43, and 100) to select distinct band subsets to mitigate the effect of randomness. As shown in Table IX, the detection performance notably degrades on most datasets. The random

TABLE II
AUC SCORES OF COMPARED HAD METHODS TRAINED AND TESTED ON THE SAME DATASET

Dataset	AUC	RX	LEBSR	AutoAD	BockNet	GTHAD	MSNet	EGSSFD	OTAD	SGLNet	Hyper SIGMA	Hyper Free	Ours
Los Angeles	AUC(P_d, P_f) ↑	0.8404	0.9325	0.9058	0.9311	0.9872	0.9807	0.9741	0.9233	0.9842	0.8935	0.7841	**0.9931**
	AUC(P_f, τ) ↓	0.0516	0.0196	0.0043	0.0121	0.0222	**0.0024**	0.0107	0.0624	0.0091	0.5036	0.0742	0.0380
	AUC(P_d, τ) ↑	0.1841	0.1672	0.0797	0.1480	0.4005	0.0792	0.1369	0.2603	0.1551	**0.7292**	0.2178	0.4103
Gulfport	AUC(P_d, P_f) ↑	0.9526	0.9726	0.9878	0.9885	0.9900	0.9919	0.9956	0.9955	0.9910	0.9912	0.9516	**0.9986**
	AUC(P_f, τ) ↓	0.0248	0.0174	0.0091	0.0210	0.0232	0.0015	0.0209	0.0958	0.0060	0.2934	0.0133	**0.0007**
	AUC(P_d, τ) ↑	0.0736	0.2657	0.2459	0.3849	0.2195	0.1333	0.3551	0.6503	0.0910	**0.7951**	0.0702	0.3751
Cat Island	AUC(P_d, P_f) ↑	0.9807	0.9428	0.9808	0.9903	0.9995	0.9782	0.9960	0.9705	0.9950	0.9691	0.9993	**0.9999**
	AUC(P_f, τ) ↓	0.0065	0.0129	0.0005	0.0005	0.0018	0.0027	0.0035	0.0227	0.0008	0.1127	0.1013	**0.0000**
	AUC(P_d, τ) ↑	0.2496	0.1119	0.1580	0.1786	0.3988	0.0695	0.2301	0.3314	0.1686	**0.8583**	0.7526	0.2827
Texas Coast	AUC(P_d, P_f) ↑	0.9946	0.9675	0.9972	0.9898	0.9948	0.9544	**0.9993**	0.9956	0.9941	0.9316	0.9495	0.9990
	AUC(P_f, τ) ↓	0.0135	0.0136	0.0011	0.0006	0.0055	**0.0001**	0.0026	0.0245	0.0005	0.1373	0.0964	0.0006
	AUC(P_d, τ) ↑	0.1178	0.1149	0.0995	0.0749	0.2125	0.0411	0.1139	0.2005	0.0440	**0.4004**	0.3554	0.1427
Gainesville	AUC(P_d, P_f) ↑	0.9513	0.9046	0.9823	0.9766	0.9836	0.9743	0.9771	0.9887	0.9719	0.8906	0.9751	**0.9951**
	AUC(P_f, τ) ↓	0.0351	0.0793	0.0109	0.0058	0.0126	**0.0021**	0.0185	0.0618	0.0057	0.2395	0.0918	0.0163
	AUC(P_d, τ) ↑	.0963	0.3199	0.1959	0.2429	0.1072	0.0759	0.2364	0.5028	0.0374	**0.5680**	0.5082	0.4931
HYDICE Urban	AUC(P_d, P_f) ↑	0.9831	0.9713	0.9961	0.9977	0.9286	0.9952	0.9947	0.9820	0.9894	0.9854	0.8771	**0.9982**
	AUC(P_f, τ) ↓	0.0345	0.0133	0.0035	0.0184	0.0539	**0.0023**	0.0141	0.1176	0.0059	0.3692	0.0248	0.0060
	AUC(P_d, τ) ↑	0.2189	0.3591	0.3131	0.4305	0.2114	0.2626	0.3830	0.6514	0.0901	**0.8118**	0.3543	0.5881
San Diego	AUC(P_d, P_f) ↑	0.9106	0.8873	0.9457	0.9321	0.9251	0.8860	0.9206	0.9055	0.8775	0.8761	0.9715	**0.9975**
	AUC(P_f, τ) ↓	0.0488	0.0111	**0.0025**	0.0030	0.0169	0.0033	0.0061	0.0420	0.0163	0.2158	0.0529	**0.0025**
	AUC(P_d, τ) ↑	0.1530	0.0609	0.0516	0.0471	0.1399	0.0241	0.0252	0.1861	0.0463	**0.4893**	0.3740	0.2774
San Diego-II	AUC(P_d, P_f) ↑	0.9491	0.9955	0.9889	0.9945	0.9926	0.9863	0.9815	0.9774	0.9780	0.9569	0.8112	**1.0000**
	AUC(P_f, τ) ↓	0.0201	0.0095	0.0142	0.0198	0.0459	0.0032	0.0080	0.1069	**0.0021**	0.1860	0.0443	0.0032
	AUC(P_d, τ) ↑	0.0478	0.1444	0.3490	0.2721	0.4646	0.1407	0.0728	0.3550	0.0126	0.5235	0.1893	**0.6740**
San Diego-III	AUC(P_d, P_f) ↑	0.9403	0.9323	0.9632	0.9882	0.9866	0.9699	0.9792	0.9787	0.9792	0.9776	0.8112	**0.9966**
	AUC(P_f, τ) ↓	0.0589	0.0069	0.0090	0.0055	0.0241	0.0021	0.0106	0.0592	0.0070	0.5096	0.0670	**0.0019**
	AUC(P_d, τ) ↑	0.1778	0.1578	0.1131	0.2230	0.3701	0.0430	0.1866	0.3931	0.0926	**0.8143**	0.2472	0.3921
Pavia	AUC(P_d, P_f) ↑	0.9538	0.9334	0.9851	0.9916	0.9908	0.9937	0.9967	0.9810	0.9964	0.8573	0.9217	**0.9993**
	AUC(P_f, τ) ↓	0.0233	0.0042	0.0017	0.0028	0.0016	**0.0001**	0.0038	0.0196	0.0016	0.2660	0.0761	0.0002
	AUC(P_d, τ) ↑	0.1343	0.0775	0.1223	0.1236	0.2183	0.0699	0.1472	0.2710	0.1194	**0.7363**	0.5053	0.2014
Cri	AUC(P_d, P_f) ↑	0.9675	0.9357	0.9710	0.9319	0.9514	0.9792	0.7681	0.9106	0.9613	0.9592	0.9045	**0.9989**
	AUC(P_f, τ) ↓	0.0236	0.0438	0.0145	0.0132	0.0023	**0.0003**	0.0614	0.1408	0.0082	0.2465	0.1661	0.0050
	AUC(P_d, τ) ↑	0.0870	0.3674	0.2630	0.1596	0.0123	0.0496	0.0952	0.3376	0.0143	**0.7065**	0.4840	0.3174
Segundo	AUC(P_d, P_f) ↑	0.9842	0.9871	0.9881	0.9626	0.9479	0.9581	0.9901	0.9796	0.9866	0.9915	0.8450	**0.9973**
	AUC(P_f, τ) ↓	0.0016	0.0033	0.0006	0.0005	0.0023	**0.0000**	0.0006	0.0071	**0.0000**	0.1311	0.1088	0.0024
	AUC(P_d, τ) ↑	0.0117	0.0616	0.0255	0.0166	0.0485	0.0070	0.0238	0.0505	0.0019	**0.3636**	0.3067	0.0559

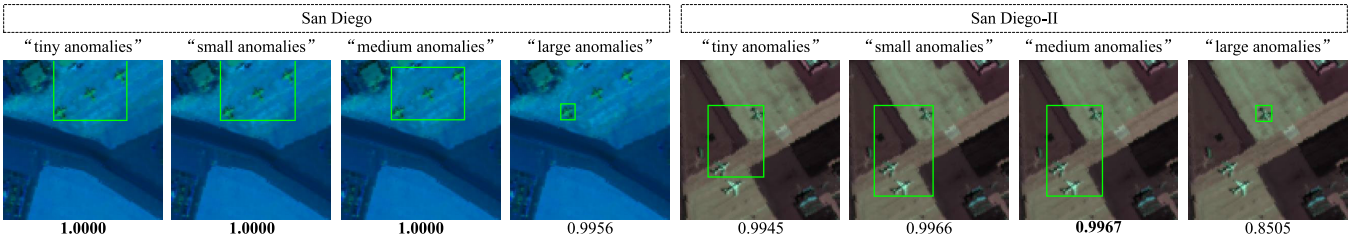

Fig. 12. Impact of the scale description in the fixed text prompt on detection performance.

band selection method introduces randomness and causes notable fluctuations in detection performance. Meanwhile, the PCA-based band reduction strategy achieves unsatisfactory detection results on several datasets. Therefore, the clustering-based band selection strategy can better align with SAM. To validate the impact of spatial information in the k-means++ method on the detection performance, we replace each pixel value of the HSI with the mean value of its $k \times k$

TABLE III

AUC(P_d, P_f) SCORES OF COMPARED HAD METHODS TRAINED ON ONE DATASET AND TESTED ON OTHER DATASETS

Train	Test	RX	LEBSR	AutoAD	BockNet	GTHAD	MSNet	EGSSFD	OTAD	SGLNet	HyperSIGMA	HyperFree	Ours
Los Angeles	Gulfport	0.9526	0.9726	0.5245	0.1981	0.5698	0.9425	0.8494	0.4561	0.4603	0.0971	0.8354	**0.9986**
	Cat Island	0.9807	0.9428	0.2960	0.7130	0.9475	0.8670	0.8979	0.8827	0.9860	0.8806	0.9468	**0.9999**
	Texas Coast	0.9946	0.9675	0.4512	0.8854	0.9937	0.9921	0.9963	0.9906	0.9956	0.2470	0.9471	**0.9990**
	Gainesville	0.9513	0.9046	0.3977	0.4928	0.4455	0.9503	0.9113	0.8192	0.8724	0.3409	0.9630	**0.9951**
	HYDICE Urban	0.9831	0.9713	0.6761	0.3522	0.3929	0.9488	0.9293	0.7868	0.4140	0.6932	0.8182	**0.9982**
	San Diego	0.9106	0.8873	0.2964	0.3534	0.8369	0.8844	0.8407	0.7116	0.8786	0.7650	0.9642	**0.9975**
	San Diego-II	0.9491	0.9955	0.1921	0.0482	0.2993	0.9974	0.9895	0.2806	0.9824	0.9644	0.7789	**1.0000**
	San Diego-III	0.9403	0.9323	0.6287	0.4966	0.7436	0.9734	0.9751	0.6890	0.8054	0.8504	0.7811	**0.9966**
	Pavia	0.9538	0.9334	0.8712	0.8220	0.8406	0.9462	0.9381	0.8861	0.7047	0.7509	0.9299	**0.9993**
	Cri	0.9675	0.9357	0.9682	0.9056	0.9820	0.9988	**0.9992**	0.9777	0.8278	0.1972	0.8915	0.9989
	Segundo	0.9842	0.9871	0.1923	0.1807	0.9533	0.9611	0.9429	0.9193	0.9821	0.7122	0.8380	**0.9973**
Gulfport	Los Angeles	0.8404	0.9325	0.6867	0.7406	0.7611	0.9455	0.8077	0.7210	0.9751	0.9077	0.7466	**0.9931**
	Cat Island	0.9807	0.9428	0.1438	0.3127	0.1801	0.9355	0.9037	0.7277	0.9761	0.8891	0.9467	**0.9999**
	Texas Coast	0.9946	0.9675	0.2856	0.9155	0.4400	0.9951	0.9786	0.7541	0.9039	0.9943	0.9390	**0.9990**
	Gainesville	0.9513	0.9046	0.6240	0.7550	0.7613	0.9530	0.8895	0.9736	0.9479	0.9381	0.9726	**0.9951**
	HYDICE Urban	0.9831	0.9713	0.6364	0.3684	0.5203	0.9763	0.9218	0.4310	0.9724	0.9665	0.8189	**0.9982**
	San Diego	0.9106	0.8873	0.2122	0.1190	0.3570	0.8901	0.6788	0.8507	0.8373	0.8910	0.9641	**0.9975**
	San Diego-II	0.9491	0.9955	0.2399	0.3105	0.3733	0.9963	0.9463	0.2591	0.9775	0.9622	0.7315	**1.0000**
	San Diego-III	0.9403	0.9323	0.6086	0.5218	0.8019	0.9584	0.9255	0.4760	0.9747	0.9023	0.7727	**0.9966**
	Pavia	0.9538	0.9334	0.8342	0.8147	0.8729	0.9635	0.9484	0.8541	0.9972	0.8763	0.8556	**0.9993**
	Cri	0.9675	0.9357	0.9591	0.8902	0.9769	0.9987	0.9974	0.9572	0.9978	0.9916	0.9047	**0.9989**
	Segundo	0.9842	0.9871	0.1129	0.3424	0.7946	0.9673	0.9539	0.5973	0.9722	0.9710	0.8163	**0.9973**
Cat Island	Los Angeles	0.8404	0.9325	0.6738	0.6094	0.7478	0.9365	0.7533	0.7253	0.9789	0.5248	0.7537	**0.9931**
	Gulfport	0.9526	0.9726	0.4611	0.3157	0.6414	0.8432	0.7925	0.9546	0.9903	0.3405	0.9509	**0.9986**
	Texas Coast	0.9946	0.9675	0.7835	0.9970	0.9979	0.9959	0.9929	0.9944	0.9936	0.9918	0.9333	**0.9990**
	Gainesville	0.9513	0.9046	0.6554	0.6029	0.6750	0.9450	0.9653	0.7059	0.9784	0.3725	0.9739	**0.9951**
	HYDICE Urban	0.9831	0.9713	0.6121	0.5039	0.6850	0.9872	0.9261	0.2565	0.9808	0.8382	0.8188	**0.9982**
	San Diego	0.9106	0.8873	0.4913	0.8724	0.8809	0.8644	0.8412	0.9006	0.8578	0.8775	0.9689	**0.9975**
	San Diego-II	0.9491	0.9955	0.4692	0.3017	0.3408	0.9964	0.9526	0.0136	0.9789	0.3671	0.7668	**1.0000**
	San Diego-III	0.9403	0.9323	0.7231	0.6554	0.7316	0.9480	0.8033	0.0423	0.9726	0.6878	0.7778	**0.9966**
	Pavia	0.9538	0.9334	0.8728	0.9012	0.8766	0.9639	0.9549	0.8045	0.9962	0.9142	0.8853	**0.9993**
	Cri	0.9675	0.9357	0.9421	0.9596	0.9815	0.9976	0.9836	0.8737	0.9970	0.9406	0.8913	**0.9989**
	Segundo	0.9842	0.9871	0.5257	0.8883	0.9858	0.9648	0.9838	0.8051	0.9836	0.9553	0.8041	**0.9973**
Texas Coast	Los Angeles	0.8404	0.9325	0.6393	0.7308	0.8498	0.9475	0.7361	0.7144	0.9790	0.3910	0.7899	**0.9931**
	Gulfport	0.9526	0.9726	0.5816	0.2055	0.5623	0.8857	0.6023	0.5014	0.6371	0.1402	0.8687	**0.9986**
	Cat Island	0.9807	0.9428	0.8169	0.9430	0.5622	0.9287	0.8413	0.9048	0.9992	0.2202	0.9731	**0.9999**
	Gainesville	0.9513	0.9046	0.7463	0.7797	0.4079	0.8920	0.8035	0.7527	0.8867	0.8728	0.9647	**0.9951**
	HYDICE Urban	0.9831	0.9713	0.7483	0.5667	0.5948	0.9160	0.8105	0.8067	0.5526	0.8404	0.8188	**0.9982**
	San Diego	0.9106	0.8873	0.7787	0.9141	0.8124	0.8658	0.6232	0.9024	0.8913	0.1378	0.9714	**0.9975**
	San Diego-II	0.9491	0.9955	0.2632	0.1742	0.3146	0.9980	0.7715	0.2753	0.9772	0.2947	0.7925	**1.0000**
	San Diego-III	0.9403	0.9323	0.6717	0.6054	0.7235	0.9248	0.6065	0.6932	0.9086	0.2698	0.7790	**0.9966**
	Pavia	0.9538	0.9334	0.8794	0.8599	0.8482	0.9655	0.8847	0.8893	0.6489	0.2729	0.9151	**0.9993**
	Cri	0.9675	0.9357	0.9676	0.9514	0.9839	0.9967	0.9884	0.9770	0.8947	0.9562	0.8936	**0.9989**
	Segundo	0.9842	0.9871	0.5316	0.6667	0.9107	0.9599	0.9779	0.9196	0.9759	0.8539	0.8345	**0.9973**
Gainesville	Los Angeles	0.8404	0.9325	0.5467	0.6495	0.7903	0.9477	0.8943	0.9027	0.9828	0.4690	0.7669	**0.9931**
	Gulfport	0.9526	0.9726	0.6782	0.6429	0.9132	0.8540	0.7891	0.6537	0.9913	0.2409	0.8276	**0.9986**
	Cat Island	0.9807	0.9428	0.2331	0.7774	0.7245	0.9440	0.8043	0.7222	0.9865	0.9281	0.9469	**0.9999**
	Texas Coast	0.9946	0.9675	0.6694	0.9912	0.9939	0.9964	0.9940	0.9902	0.9954	**0.9992**	0.9503	0.9990
	HYDICE Urban	0.9831	0.9713	0.6760	0.5274	0.2770	0.9177	0.9198	0.7843	0.9725	0.9659	0.8184	**0.9982**
	San Diego	0.9106	0.8873	0.3655	0.3233	0.8353	0.8899	0.2549	0.9004	0.8626	0.8953	0.9715	**0.9975**
	San Diego-II	0.9491	0.9955	0.2230	0.1815	0.0292	0.9952	0.9844	0.2681	0.9768	0.2394	0.8046	**1.0000**
	San Diego-III	0.9403	0.9323	0.6363	0.5892	0.0835	0.9182	0.8813	0.6875	0.9705	0.5801	0.7629	**0.9966**
	Pavia	0.9538	0.9334	0.8839	0.8434	0.7993	0.9810	0.9201	0.8842	0.9941	0.9229	0.9296	**0.9993**
	Cri	0.9675	0.9357	0.9497	0.9399	0.6462	0.9985	0.9970	0.9710	0.9972	0.9800	0.8868	**0.9989**
	Segundo	0.9842	0.9871	0.2192	0.3554	0.9281	0.9796	0.9627	0.9365	0.9810	0.9639	0.8425	**0.9973**
HYDICE Urban	Los Angeles	0.8404	0.9325	0.5496	0.7418	0.7566	0.9587	0.6885	0.8424	0.9772	0.9683	0.7433	**0.9931**
	Gulfport	0.9526	0.9726	0.6237	0.8731	0.7893	0.9731	0.7830	0.6302	0.9792	0.3948	0.9326	**0.9986**
	Cat Island	0.9807	0.9428	0.2533	0.7195	0.9392	0.9236	0.1955	0.4914	0.9875	0.6888	0.8936	**0.9999**
	Texas Coast	0.9946	0.9675	0.4026	0.9086	0.6976	0.9948	0.5604	0.8549	0.9922	0.9921	0.9416	**0.9990**
	Gainesville	0.9513	0.9046	0.5265	0.7255	0.5529	0.9273	0.3974	0.4452	0.9654	0.5169	0.9712	**0.9951**
	San Diego	0.9106	0.8873	0.4876	0.4575	0.5723	0.8701	0.5453	0.3421	0.8362	0.7661	0.9740	**0.9975**
	San Diego-II	0.9491	0.9955	0.3967	0.4412	0.2656	0.9932	0.7486	0.3828	0.9681	0.9910	0.7553	**1.0000**
	San Diego-III	0.9403	0.9323	0.7030	0.7834	0.8214	0.9849	0.4751	0.7133	0.9659	0.9768	0.7797	**0.9966**
	Pavia	0.9538	0.9334	0.1292	0.5385	0.3043	0.9633	0.4638	0.7372	0.9957	0.7553	0.8850	**0.9993**
	Cri	0.9675	0.9357	0.9326	0.9224	0.8760	0.9959	0.5365	0.9742	0.9968	0.9884	0.8778	**0.9989**
	Segundo	0.9842	0.9871	0.3170	0.7379	0.8499	0.9601	0.6493	0.7956	0.9715	0.9330	0.8090	**0.9973**
San Diego	Los Angeles	0.8404	0.9325	0.6162	0.6568	0.6227	0.9470	0.8704	0.5910	0.9752	0.4785	0.7303	**0.9931**
	Gulfport	0.9526	0.9726	0.3992	0.5082	0.1801	0.9457	0.8967	0.2899	0.9923	0.1830	0.8990	**0.9986**
	Cat Island	0.9807	0.9428	0.8220	0.8667	0.4476	0.9401	0.9240	0.8001	0.9910	0.9729	0.9464	**0.9999**
	Texas Coast	0.9946	0.9675	0.9781	0.9873	0.4973	0.9940	0.9807	0.5977	0.9955	0.9983	0.9459	**0.9990**
	Gainesville	0.9513	0.9046	0.6644	0.6504	0.2354	0.9493	0.8697	0.5398	0.9740	0.7355	0.9657	**0.9951**
	HYDICE Urban	0.9831	0.9713	0.7426	0.2534	0.3665	0.9124	0.8679	0.3949	0.9633	0.9478	0.8186	**0.9982**
	San Diego-II	0.9491	0.9955	0.2281	0.1239	0.2809	0.9945	0.9400	0.1847	0.9811	0.1967	0.7729	**1.0000**
	San Diego-III	0.9403	0.9323	0.6601	0.3903	0.6982	0.8969	0.8565	0.5174	0.9756	0.6526	0.7919	**0.9966**
	Pavia	0.9538	0.9334	0.8715	0.7835	0.8435	0.9775	0.9521	0.8482	0.9944	0.9448	0.8781	**0.9993**
	Cri	0.9675	0.9357	0.9705	0.8859	0.9767	0.9981	0.9932	0.9712	0.9979	0.9819	0.9150	**0.9989**
	Segundo	0.9842	0.9871	0.5609	0.9411	0.5733	0.9676	0.9625	0.2319	0.9785	0.9674	0.8209	**0.9973**
San Diego-II	Los Angeles	0.8404	0.9325	0.2972	0.8431	0.5360	0.9517	0.9622	0.4289	0.9809	0.8834	0.7732	**0.9931**
	Gulfport	0.9526	0.9726	0.8700	0.8295	0.6596	0.9314	0.9397	0.9522	0.4575	0.4828	0.9598	**0.9986**
	Cat Island	0.9807	0.9428	0.5000	0.6801	0.1805	0.9103	0.9337	0.0471	0.9938	0.9996	0.9464	**0.9999**
	Texas Coast	0.9946	0.9675	0.2309	0.6145	0.8434	0.9878	0.9928	0.1473	0.9850	0.8004	0.9399	**0.9990**
	Gainesville	0.9513	0.9046	0.3188	0.2377	0.7258	0.9660	0.9499	0.6325	0.4998	0.9934	0.9731	**0.9951**
	HYDICE Urban	0.9831	0.9713	0.3012	0.2510	0.5323	0.8178	0.9430	0.4676	0.0703	0.9774	0.8187	**0.9982**
	San Diego	0.9106	0.8873	0.4589	0.5423	0.4789	0.9165	0.5482	0.1517	0.8078	0.9653	0.9688	**0.9975**
	San Diego-III	0.9403	0.9323	0.4487	0.7154	0.7549	0.9711	0.9843	0.5790	0.7675	0.7194	0.7997	**0.9966**
	Pavia	0.9538	0.9334	0.0490	0.0775	0.0859	0.9443	0.9569	0.1096	0.3217	0.9663	0.8852	**0.9993**
	Cri	0.9675	0.9357	0.1892	0.1167	0.9656	0.9964	0.9961	0.9648	0.9929	0.8938	0.8955	**0.9989**
	Segundo	0.9842	0.9871	0.3160	0.4164	0.5774	0.9254	0.9535	0.1059	0.9803	0.9519	0.8133	**0.9973**
San Diego-III	Los Angeles	0.8404	0.9325	0.6453	0.9440	0.7740	0.9494	0.9692	0.4912	0.9819	0.8125	0.7689	**0.9931**
	Gulfport	0.9526	0.9726	0.6703	0.9932	0.9424	0.9073	0.8885	0.9307	0.9894	0.0988	0.9182	**0.9986**
	Cat Island	0.9807	0.9428	0.0742	0.5672	0.6479	0.8529	0.9207	0.1424	0.9872	0.9767	0.9469	**0.9999**
	Texas Coast	0.9946	0.9675	0.5434	0.9744	0.9873	0.9884	0.9897	0.5923	0.9942	0.6489	0.9497	**0.9990**
	Gainesville	0.9513	0.9046	0.6488	0.7802	0.9033	0.8712	0.9810	0.5961	0.9675	0.7416	0.9644	**0.9951**
	HYDICE Urban	0.9831	0.9713	0.3720	0.4151	0.5807	0.8939	0.9862	0.6816	0.9768	0.9607	0.8189	**0.9982**
	San Diego	0.9106	0.8873	0.5716	0.4219	0.7509	0.8833	0.6108	0.2373	0.8660	0.9729	0.9690	**0.9975**
	San Diego-II	0.9491	0.9955	0.0912	0.4417	0.1622	0.9916	0.9889	0.2276	0.9845	0.8051	0.7758	**1.0000**
	Pavia	0.9538	0.9334	0.2775	0.7411	0.3964	0.9482	0.9329	0.7919	0.9970	0.9041	0.8778	**0.9993**
	Cri	0.9675	0.9357	0.8797	0.7691	0.8491	0.9966	0.9984	0.9143	0.9972	0.9471	0.8864	**0.9989**
	Segundo	0.9842	0.9871	0.4127	0.6663	0.8615	0.9441	0.9335	0.3182	0.9791	0.8699	0.8130	**0.9973**
Pavia	Los Angeles	0.8404	0.9325	0.5894	0.7329	0.8795	0.9714	0.8864	0.7522	0.9708	0.6635	0.7600	**0.9931**
	Gulfport	0.9526	0.9726	0.0619	0.5776	0.6787	0.9520	0.9356	0.6047	0.9816	0.6288	0.9339	**0.9986**
	Cat Island	0.9807	0.9428	0.3308	0.9441	0.9566	0.9396	0.9695	0.3987	0.9957	0.9642	0.9469	**0.9999**
	Texas Coast	0.9946	0.9675	0.4400	0.9984	0.9980	0.9935	0.9848	**0.9990**	0.9951	0.9851	0.9507	**0.9990**
	Gainesville	0.9513	0.9046	0.3524	0.7258	0.6965	0.9775	0.9245	0.8243	0.9814	0.8679	0.9637	**0.9951**
	HYDICE Urban	0.9831	0.9713	0.4743	0.6442	0.3346	0.9035	0.9963	0.5556	0.9889	0.9359	0.8186	**0.9982**
	San Diego	0.9106	0.8873	0.4548	0.6109	0.7400	0.9063	0.3098	0.4527	0.8617	0.8981	0.9714	**0.9975**
	San Diego-II	0.9491	0.9955	0.4147	0.5957	0.5398	0.9949	0.9462	0.5014	0.9784	0.4834	0.7881	**1.0000**
	San Diego-III	0.9403	0.9323	0.7423	0.7450	0.9539	0.9565	0.8897	0.7455	0.9695	0.7005	0.7562	**0.9966**
	Cri	0.9675	0.9357	0.9196	0.9262	0.9077	0.9971	0.9527	0.9495	0.9975	0.9486	0.8951	**0.9989**
	Segundo	0.9842	0.9871	0.3292	0.9908	0.9718	0.9535	0.9549	0.9915	0.9852	0.9768	0.8394	**0.9973**
Cri	Los Angeles	0.8404	0.9325	0.4658	0.6842	0.8613	0.9515	0.9142	0.8190	0.9480	0.5890	0.7548	**0.9931**
	Gulfport	0.9526	0.9726	0.1563	0.5189	0.3833	0.9653	0.8141	0.2785	0.9783	0.2015	0.9348	**0.9986**
	Cat Island	0.9807	0.9428	0.0257	0.3303	0.0509	0.9369	0.9628	0.4681	0.9960	0.9552	0.9990	**0.9999**
	Texas Coast	0.9946	0.9675	0.0012	0.6203	0.2102	0.9893	0.9955	0.5378	0.8943	0.8877	0.9420	**0.9990**
	Gainesville	0.9513	0.9046	0.0441	0.4143	0.0893	0.9607	0.8302	0.3445	0.4155	0.6095	0.9849	**0.9951**
	HYDICE Urban	0.9831	0.9713	0.0352	0.7848	0.2806	0.9773	0.9797	0.9114	0.9938	0.9670	0.8186	**0.9982**
	San Diego	0.9106	0.8873	0.1519	0.2525	0.0979	0.8734	0.8005	0.0967	0.8817	0.6601	0.9689	**0.9975**
	San Diego-II	0.9491	0.9955	0.5813	0.5721	0.6930	0.9994	0.9837	0.5900	0.9861	0.6215	0.7765	**1.0000**
	San Diego-III	0.9403	0.9323	0.5071	0.7174	0.6632	0.9747	0.9076	0.8993	0.9766	0.6306	0.7877	**0.9966**
	Pavia	0.9538	0.9334	0.1426	0.6662	0.1409	0.9062	0.9822	0.8540	0.9933	0.9319	0.8706	**0.9993**
	Segundo	0.9842	0.9871	0.0150	0.1619	0.0413	0.9840	0.9059	0.4273	0.9834	0.9767	0.8135	**0.9973**
Segundo	Los Angeles	0.8404	0.9325	0.7019	0.7429	0.7140	0.9317	0.5046	0.6251	0.6476	0.6373	0.7855	**0.9931**
	Gulfport	0.9526	0.9726	0.6214	0.3581	0.6299	0.8687	0.7340	0.5949	0.9802	0.3008	0.8448	**0.9986**
	Cat Island	0.9807	0.9428	0.9606	0.9675	0.9671	0.9496	0.5125	0.9727	0.9996	0.9684	0.9469	**0.9999**
	Texas Coast	0.9946	0.9675	0.9988	0.9951	0.9989	0.9877	0.5579	**0.9994**	0.9922	0.9992	0.9540	0.9990
	Gainesville	0.9513	0.9046	0.8133	0.7817	0.7502	0.9193	0.5976	0.7926	0.9697	0.6962	0.9747	**0.9951**
	HYDICE Urban	0.9831	0.9713	0.7858	0.6866	0.7538	0.8817	0.4829	0.8546	0.9957	0.9739	0.8183	**0.9982**
	San Diego	0.9106	0.8873	0.8191	0.8829	0.8769	0.8684	0.4908	0.8964	0.9088	0.8957	0.9638	**0.9975**
	San Diego-II	0.9491	0.9955	0.2829	0.0815	0.3070	0.9970	0.4261	0.2787	0.9863	0.4982	0.8058	**1.0000**
	San Diego-III	0.9403	0.9323	0.7049	0.6190	0.7206	0.9124	0.4328	0.7036	0.9700	0.8197	0.7855	**0.9966**
	Pavia	0.9538	0.9334	0.9032	0.8950	0.8877	0.9208	0.4711	0.9062	0.9961	0.9113	0.9374	**0.9993**
	Cri	0.9675	0.9357	0.9527	0.9431	0.9657	0.9983	0.4779	0.9662	0.5814	0.9521	0.8916	**0.9989**

TABLE IV
EFFECT OF SAM ON DETECTION PERFORMANCE

Variants	Los Angeles	Gulfport	Cat Island	Texas Coast	Gainesville	HYDICE Urban	San Diego	San Diego-II	San Diego-III	Pavia	Cri	Segundo
w/o SAM	0.9817	0.9401	0.9893	0.9984	0.9605	0.9904	0.9775	0.9996	0.9950	0.9594	0.9976	0.9926
Ours	**0.9931**	**0.9986**	**0.9999**	**0.9990**	**0.9951**	**0.9982**	**0.9975**	**1.0000**	**0.9966**	**0.9993**	**0.9989**	**0.9973**

TABLE V
EFFECT OF BOX PROMPT IN SAM ON DETECTION PERFORMANCE

Variants	Los Angeles	Gulfport	Cat Island	Texas Coast	Gainesville	HYDICE Urban	San Diego	San Diego-II	San Diego-III	Pavia	Cri	Segundo
SAM w box prompt	0.9925	0.9785	**0.9999**	**0.9990**	0.9903	0.9973	0.9965	**1.0000**	**0.9980**	0.9899	**0.9989**	**0.9975**
Ours	**0.9931**	**0.9986**	**0.9999**	**0.9990**	**0.9951**	**0.9982**	**0.9975**	**1.0000**	0.9966	**0.9993**	**0.9989**	0.9973

TABLE VI
EFFECT OF MODEL PARAMETER SIZE ON DETECTION PERFORMANCE

SAM	Grounding DINO	Los Angeles	Gulfport	Cat Island	Texas Coast	Gainesville	HYDICE Urban	San Diego	San Diego-II	San Diego-III	Pavia	Cri	Segundo
Large	Tiny	0.9894	0.9466	**0.9999**	0.9986	0.9613	0.9793	0.9954	0.9999	0.9874	0.9884	0.9969	0.9965
Huge	Tiny	0.9913	0.9846	0.9737	0.9987	0.9512	0.9879	0.9946	**1.0000**	0.9935	0.9901	0.9985	0.9869
Base	Base	0.9297	0.9927	0.9764	**0.9990**	0.9844	0.9980	**0.9975**	**1.0000**	**0.9967**	**0.9993**	**0.9991**	**0.9973**
Large	Base	0.8087	0.9793	0.9736	0.9986	0.9687	0.9771	0.9954	0.9999	0.9875	0.9884	0.9969	0.9965
Huge	Base	0.4427	0.9472	0.9777	0.9987	0.6237	0.9875	0.9946	**1.0000**	0.9935	0.9901	0.9985	0.9869
Ours (Base)	Ours (Tiny)	**0.9931**	**0.9986**	**0.9999**	**0.9990**	**0.9951**	**0.9982**	**0.9975**	**1.0000**	0.9966	**0.9993**	0.9989	**0.9973**

TABLE VII
EFFECT OF THE POSTPROCESSING STRATEGY ON DETECTION PERFORMANCE

Variants	Los Angeles	Gulfport	Cat Island	Texas Coast	Gainesville	HYDICE Urban	San Diego	San Diego-II	San Diego-III	Pavia	Cri	Segundo
w/o $\mathbf{A}_{SAM}$	0.9817	0.9401	0.9893	0.9984	0.9605	0.9904	0.9775	0.9996	0.9950	0.9594	0.9976	0.9926
w/o $\mathbf{A}_{RX}$	0.9413	**0.9989**	0.9998	0.9986	0.9762	0.9940	**0.9976**	0.9985	0.9956	**0.9995**	0.9971	0.9739
w/o $\mathbf{M}$	0.9927	0.9986	**0.9999**	**0.9990**	**0.9952**	**0.9993**	0.9962	**1.0000**	0.9959	0.9993	0.9982	0.9954
w/o $\mathbf{A}_{RX}$ and $\mathbf{M}$	0.9413	**0.9989**	0.9998	0.9986	0.9932	0.9954	0.9975	0.9985	0.9945	**0.9995**	0.9946	0.9685
Ours	**0.9931**	0.9986	**0.9999**	**0.9990**	0.9951	0.9982	0.9975	**1.0000**	**0.9966**	0.9993	**0.9989**	**0.9973**

neighboring window. The value of k is set to $\{3, 5, 7, 9\}$. The experimental results from Table X show that detection performance remains unchanged on most datasets when only spectral information is adopted in the clustering method. For the datasets with tiny targets such as HYDICE Urban and Pavia, the detection performance degrades when introducing the spatial information. That is because the features of tiny targets tend to be submerged by the surrounding background features. Therefore, we only use the spectral distance metric in the k-means++ method.

2) Effect of the Text Prompt: Due to the domain shift, the AUC(P_d, P_f) scores achieved by SAM are much lower on all datasets, as shown in Table XI. Our method proposes an ensemble text prompt strategy. Using either ATP or FTP can improve detection performance on part of the datasets. In contrast, integrating both prompts yields performance gains on all datasets and demonstrates better scene adaptability. We also test the effect of different text prompts on the detection performance. As shown in Fig. 14, when different prompts are applied to our method, the Grounding DINO generates different boxes (colored in green) that locate the target or background regions. By employing the text prompt generation module, the proposed method effectively identifies target regions and achieves the highest AUC (P_d, P_f) scores on most datasets.

3) Effect of the Point Prompt: By adding the point prompt to SAM, the detection performance is also notably improved on most datasets (see Table XI). Moreover, as shown in

TABLE VIII

AUC(P_d, P_f) SCORES OF COMPARED HAD METHODS TRAINED ON THE TWO DATASETS WITH BLURRED TARGET BOUNDARIES AND TESTED ON OTHER DATASETS

Train	Test	RX	LEBSR	AutoAD	BockNet	GTHAD	MSNet	EGSSFD	OTAD	SGLNet	HyperSIGMA	HyperFree	Ours
Bay Champagne	Los Angeles	0.8404	0.9325	0.6146	0.6751	0.2385	0.9303	0.9423	0.3794	0.9831	0.5307	0.7817	**0.9931**
	Gulfport	0.9526	0.9726	0.3720	0.2701	0.2918	0.9607	0.8432	0.7231	0.9911	0.1296	0.9250	**0.9986**
	Cat Island	0.9807	0.9428	0.0086	0.3465	0.5758	0.9351	0.9469	0.8918	0.9843	0.9084	0.9732	**0.9999**
	Texas Coast	0.9946	0.9675	0.2570	0.9151	0.6005	0.9946	0.9983	0.5665	0.9941	0.9970	0.9507	**0.9990**
	Gainesville	0.9513	0.9046	0.2184	0.5069	0.2781	0.9501	0.9570	0.2267	0.9725	0.7528	0.9753	**0.9951**
	HYDICE Urban	0.9831	0.9713	0.5028	0.4690	0.3909	0.9055	0.9468	0.0290	0.3608	0.9100	0.8191	**0.9982**
	San Diego	0.9106	0.8873	0.1876	0.2093	0.2534	0.8647	0.8370	0.8995	0.8592	0.4339	0.9714	**0.9975**
	San Diego-II	0.9491	0.9955	0.2687	0.4763	0.0621	0.9944	0.9588	0.0925	0.9790	0.2927	0.7968	**1.0000**
	San Diego-III	0.9403	0.9323	0.5308	0.7073	0.2512	0.9550	0.9410	0.2467	0.9736	0.6407	0.8088	**0.9966**
	Pavia	0.9538	0.9334	0.8372	0.7868	0.7908	0.9935	0.9469	0.7859	0.6894	0.9004	0.9225	**0.9993**
	Cri	0.9675	0.9357	0.9547	0.9213	0.8871	0.9921	0.9830	0.8997	0.9960	0.8400	0.8931	**0.9989**
	Segundo	0.9842	0.9871	0.1485	0.3405	0.5081	0.9634	0.9773	0.2325	0.9774	0.9472	0.8327	**0.9973**
	Bay Champagne	0.9998	0.9960	0.9990	0.9814	0.9997	0.9995	0.9978	0.9970	**1.0000**	0.9610	0.9837	**1.0000**
	HyMap	0.9028	0.8954	0.7237	0.9515	0.7587	0.9113	0.9738	0.1900	0.9774	0.9603	0.9806	**0.9960**
HyMap	Los Angeles	0.8404	0.9325	0.8020	0.8535	0.8686	0.9361	0.4714	0.7468	0.9800	0.5507	0.7763	**0.9931**
	Gulfport	0.9526	0.9726	0.6407	0.7595	0.7144	0.8940	0.5663	0.5478	0.9857	0.5690	0.8272	**0.9986**
	Cat Island	0.9807	0.9428	0.6695	0.7240	0.9229	0.9357	0.3398	0.8287	0.9926	0.9618	0.9468	**0.9999**
	Texas Coast	0.9946	0.9675	0.9929	0.9845	0.9970	0.9961	0.6119	**0.9990**	0.9919	0.9972	0.9509	**0.9990**
	Gainesville	0.9513	0.9046	0.7970	0.8301	0.4954	0.8722	0.5359	0.8119	0.9688	0.8432	0.9722	**0.9951**
	HYDICE Urban	0.9831	0.9713	0.6455	0.6023	0.5883	0.9795	0.2278	0.7979	0.9791	0.9216	0.8183	**0.9982**
	San Diego	0.9106	0.8873	0.6328	0.6069	0.3384	0.8486	0.4619	0.4408	0.8675	0.9036	0.9640	**0.9975**
	San Diego-II	0.9491	0.9955	0.1915	0.3808	0.3439	0.9946	0.3833	0.3320	0.9786	0.3939	0.7947	**1.0000**
	San Diego-III	0.9403	0.9323	0.6652	0.7963	0.7610	0.9855	0.4693	0.7277	0.9754	0.6900	0.7810	**0.9966**
	Pavia	0.9538	0.9334	0.8827	0.9308	0.8635	0.9594	0.4500	0.9008	0.9986	0.9079	0.9297	**0.9993**
	Cri	0.9675	0.9357	0.9317	0.8553	0.9734	0.9969	0.4814	0.9793	0.9950	0.9792	0.8815	**0.9989**
	Segundo	0.9842	0.9871	0.6756	0.8510	0.9690	0.9760	0.6013	0.9884	0.9782	0.9737	0.8431	**0.9973**
	Bay Champagne	0.9998	0.9960	0.9611	0.8828	0.9631	0.9529	0.1757	0.9324	0.9999	0.9787	0.9875	**1.0000**
	HyMap	0.9028	0.8954	0.9224	0.9236	0.9800	0.9825	0.9868	0.9614	0.9678	0.9662	0.9823	**0.9960**

TABLE IX

EFFECT OF THE BAND SELECTION STRATEGY

Variants	Los Angeles	Gulfport	Cat Island	Texas Coast	Gainesville	HYDICE Urban	San Diego	San Diego-II	San Diego-III	Pavia	Cri	Segundo
PCA	0.8780	0.9986	0.9483	**0.9991**	0.9515	0.9839	0.9037	**1.0000**	**0.9981**	0.9990	0.9989	**0.9975**
Random selection (seed=1)	0.8746	**0.9989**	0.9332	0.9055	0.8236	0.9976	0.9449	0.9997	0.9000	0.9684	0.9985	0.9965
Random selection (seed=43)	**0.9931**	0.9947	0.9574	0.9990	0.9493	0.9905	0.8954	0.9966	0.9921	0.8191	0.9979	0.9901
Random selection (seed=100)	0.9167	0.9979	0.9759	0.9986	0.9565	**0.9982**	0.9972	0.9969	0.9974	0.9603	**0.9991**	0.9898
Ours (k-means++)	**0.9931**	0.9986	**0.9999**	0.9990	**0.9951**	**0.9982**	**0.9975**	**1.0000**	0.9966	**0.9993**	0.9989	0.9973

TABLE X

EFFECT OF ADOPTING SPATIAL INFORMATION IN BAND SELECTION

Variants	Los Angeles	Gulfport	Cat Island	Texas Coast	Gainesville	HYDICE Urban	San Diego	San Diego-II	San Diego-III	Pavia	Cri	Segundo
3×3	0.9931	0.9986	0.9999	0.9990	0.9948	0.9975	0.9976	1.0000	0.9962	0.9993	0.9989	0.9973
5×5	0.9931	0.9986	0.9999	0.9990	0.9948	0.9975	0.9976	1.0000	0.9957	0.9150	0.9989	0.9973
7×7	0.9931	0.9986	0.9999	0.9990	0.9948	0.9975	0.9976	1.0000	0.9960	0.9181	0.9989	0.9973
9×9	0.9931	0.9986	0.9999	0.9990	0.9948	0.9975	0.9976	1.0000	0.9958	0.9181	0.9989	0.9973
Ours	0.9931	0.9986	0.9999	0.9990	0.9951	0.9982	0.9975	1.0000	0.9966	0.9993	0.9989	0.9973

Fig. 15, the point prompts can accurately locate the anomaly pixels (colored in green) and the background pixels (colored in red). The selected background points are outside the target masks generated by corresponding text prompts. We also investigate the impact of the anomaly point prompt and background point prompt on the detection performance, respectively. As shown in Table XII, the detection performance degrades on most datasets using either anomaly point prompts or background point prompts. In contrast, the anomaly prompt provides more contribution to the detection performance.

TABLE XI
AUC(P_d, P_f) SCORES OF DIFFERENT VARIANTS

Foundation model	PP	ATP	FTP	Los Angeles	Gulfport	Cat Island	Texas Coast	Gainesville	HYDICE Urban	San Diego	San Diego-II	San Diego-III	Pavia	Cri	Segundo
SAM	✗	✗	✗	0.4565	0.4949	0.9451	0.9679	0.8832	0.9145	0.9087	0.9822	0.5767	0.9260	0.9783	0.8896
SAM	✓	✗	✗	0.9529	0.9831	0.9762	**0.9990**	0.8979	**0.9994**	0.9123	0.8608	0.9657	**0.9994**	0.9981	0.9957
SAM	✗	✓	✗	**0.9957**	0.9199	0.9286	0.9981	0.7438	0.9722	0.9886	0.9959	0.8118	0.9848	0.9965	0.9959
SAM	✗	✗	✓	0.9132	0.9188	**0.9999**	0.9982	0.9412	0.9747	0.9967	0.9998	0.9922	0.7777	0.9973	0.9960
SAM	✓	✗	✓	0.8660	0.9912	**0.9999**	**0.9990**	**0.9951**	**0.9982**	**0.9975**	**1.0000**	**0.9966**	0.9149	0.9989	**0.9973**
SAM	✓	✓	✗	0.9931	**0.9986**	0.9382	**0.9990**	0.8517	0.9981	0.9805	0.9956	0.9788	0.9993	**0.9990**	**0.9973**
SAM	✗	✓	✓	**0.9957**	0.9199	**0.9999**	0.9981	0.9412	0.9747	0.9967	0.9998	0.9922	0.9848	0.9973	0.9960
SAM	✓	✓	✓	0.9931	**0.9986**	**0.9999**	**0.9990**	**0.9951**	0.9982	**0.9975**	**1.0000**	**0.9966**	0.9993	0.9989	**0.9973**

TABLE XII
EFFECT OF POINT PROMPT TYPE ON DETECTION PERFORMANCE

Variants	Los Angeles	Gulfport	Cat Island	Texas Coast	Gainesville	HYDICE Urban	San Diego	San Diego-II	San Diego-III	Pavia	Cri	Segundo
Anomaly prompt	0.9928	0.9977	**0.9999**	**0.9990**	0.9769	0.9579	**0.9975**	**1.0000**	0.9865	0.9988	0.9893	0.9700
Background prompt	0.9451	0.8464	0.8542	0.9598	0.9042	0.9383	0.9671	0.9598	0.9333	0.8482	0.9923	0.9950
Ours	**0.9931**	**0.9986**	**0.9999**	**0.9990**	**0.9951**	**0.9982**	**0.9975**	**1.0000**	**0.9966**	**0.9993**	**0.9989**	**0.9973**

TABLE XIII
EFFECT OF BAND SELECTION APPROACH FOR RX ON THE DETECTION PERFORMANCE

Variants	Los Angeles	Gulfport	Cat Island	Texas Coast	Gainesville	HYDICE Urban	San Diego	San Diego-II	San Diego-III	Pavia	Cri	Segundo
k-means++	0.9917	0.9773	**1.0000**	0.9989	**0.9965**	0.9973	0.9909	**1.0000**	**0.9967**	**0.9996**	**0.9990**	0.9968
The whole HSI	**0.9931**	0.9847	0.9999	0.9987	0.9871	0.9851	0.9974	0.9974	0.9802	0.9982	0.9987	0.9938
PCA (ours)	**0.9931**	**0.9986**	0.9999	**0.9990**	0.9951	**0.9982**	**0.9975**	**1.0000**	0.9966	0.9993	0.9989	**0.9973**

TABLE XIV
AUC(P_d, P_f) SCORES OF DIFFERENT TYPES OF PROMPTS ON THE HYPERFREE MODEL

Foundation model	PP	ATP	FTP	Los Angeles	Gulfport	Cat Island	Texas Coast	Gainesville	HYDICE Urban	San Diego	San Diego-II	San Diego-III	Pavia	Cri	Segundo
HyperFree	✗	✗	✗	0.7601	0.9173	0.9730	0.9490	0.9652	0.8187	0.9714	0.7755	0.7770	0.8856	0.8790	0.8217
HyperFree	✓	✗	✗	0.8910	0.9836	0.9827	**0.9992**	0.9220	0.9641	0.8881	0.8145	0.9860	0.9803	0.9066	0.7635
HyperFree	✗	✓	✗	**0.9940**	0.9312	0.8784	0.9982	0.7767	0.9159	0.9762	0.9863	0.8619	0.9827	0.9914	0.9936
HyperFree	✗	✗	✓	0.9153	0.9296	**0.9999**	0.9976	0.9305	0.9537	0.9969	0.9988	0.9934	0.7123	0.9817	0.9937
HyperFree	✓	✗	✓	0.9107	0.9785	**0.9999**	**0.9992**	0.9736	0.9852	0.9913	0.9999	**0.9973**	0.8629	0.9423	0.7860
HyperFree	✓	✓	✗	0.9931	0.9930	0.9290	**0.9992**	0.8831	0.9803	0.9585	0.9881	0.9670	0.9809	0.9359	0.7841
HyperFree	✗	✓	✓	**0.9940**	0.9312	**0.9999**	0.9982	0.9305	0.9537	0.9969	0.9988	0.9934	0.9827	0.9817	0.9937
HyperFree	✓	✓	✓	0.9931	0.9930	**0.9999**	**0.9992**	0.9736	0.9852	0.9913	0.9999	**0.9973**	0.9809	0.9423	0.7860
SAM	✓	✓	✓	0.9931	**0.9986**	**0.9999**	0.9990	**0.9951**	**0.9982**	**0.9975**	**1.0000**	0.9966	**0.9993**	**0.9989**	**0.9973**

4) Effect of the Band Selection for RX: The proposed method uses the RX algorithm to obtain the initial anomaly maps for generating point prompts. We investigate the effect of the band selection approach for RX on the detection performance, including k-means++, the whole HSI, and PCA. As shown in Table XIII, when PCA is performed on the HSIs and preserves five principal components as the input of RX, our method can achieve the best performance on most datasets.

5) Effect of Foundation Model Selection: We replace SAM with the HyperFree model to investigate the effect of the foundation model on the detection performance. Since HyperFree can automatically align with the number of HSI bands, we perform the dual prompts on this model. As shown in Table XIV, the detection performance obtained by HyperFree is also improved by using the two proposed prompt strategies, which further validates their effectiveness. However, the AUC scores of HyperFree on the Gulfport, Gainesville,

TABLE XV
IMPACT OF r ON OUR METHOD

Variants	Los Angeles	Gulfport	Cat Island	Texas Coast	Gainesville	HYDICE Urban	San Diego	San Diego-II	San Diego-III	Pavia	Cri	Segundo
r=1	0.9831	0.6413	**1.0000**	**0.9993**	0.7441	0.9460	0.9912	0.9943	0.9771	0.9504	0.9849	0.9917
r=3	0.9916	0.9923	**1.0000**	0.9990	0.9939	0.9958	0.9944	**1.0000**	0.9920	**0.9995**	**0.9990**	0.9972
r=10	0.9890	0.9913	0.9999	0.9969	0.9910	0.9969	**0.9975**	0.9993	**0.9974**	0.9985	0.9989	0.9944
Ours (r=5)	**0.9931**	**0.9986**	0.9999	0.9990	**0.9951**	**0.9982**	**0.9975**	**1.0000**	0.9966	0.9993	0.9989	**0.9973**

TABLE XVI
IMPACT OF α ON OUR METHOD

α	Los Angeles	Gulfport	Cat Island	Texas Coast	Gainesville	HYDICE Urban	San Diego	San Diego-II	San Diego-III	Pavia	Cri	Segundo
0.01	**0.9931**	**0.9986**	**0.9999**	**0.9990**	0.9781	0.9966	**0.9977**	**1.0000**	**0.9968**	**0.9993**	0.9984	**0.9974**
0.05	**0.9931**	**0.9986**	**0.9999**	**0.9990**	0.9931	0.9973	0.9976	**1.0000**	0.9967	**0.9993**	0.9988	**0.9974**
0.2	**0.9931**	**0.9986**	**0.9999**	**0.9990**	0.9960	0.9987	0.9973	**1.0000**	0.9965	**0.9993**	**0.9990**	0.9972
0.5	0.9930	**0.9986**	**0.9999**	**0.9990**	**0.9962**	**0.9992**	0.9968	**1.0000**	0.9962	**0.9993**	0.9988	0.9965
Ours (0.1)	**0.9931**	**0.9986**	**0.9999**	**0.9990**	0.9951	0.9982	0.9975	**1.0000**	0.9966	**0.9993**	0.9989	0.9973

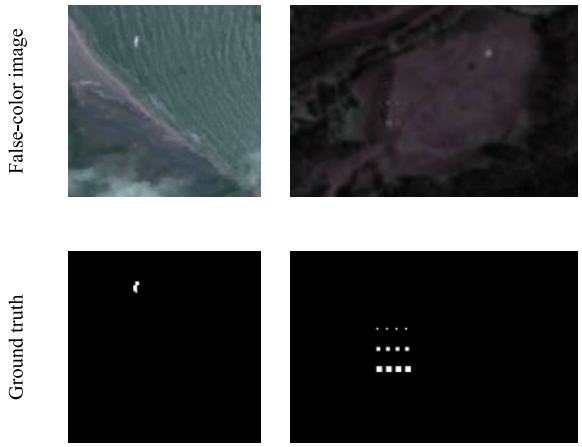

Fig. 13. False-color and Ground truth images of the two public datasets with blurred target boundaries.

HYDICE Urban, San Diego, San Diego-II, Pavia, Cri, and Segundo datasets are inferior to those of our method. Thereby, our method is more robust to different scenes. In fact, the foundation models such as SAM and HyperFree achieve much lower AUC scores on all datasets without prompts due to the semantic poverty on the downstream task. When both proposed prompts are applied to them, the detection performance is remarkably improved on all datasets.

E. Parameter Analysis

1) Impact of r: r denotes the channel dimension of the original HSI after performing PCA, which influences the anomaly map that is used for point prompting. An excessively small value of r leads to the loss of spectral information, while an overly large r introduces noise interference. Hence, an appropriate value of r is necessary for RX to obtain the anomaly map. Experimental results are shown in Table XV. The value of r is set to $\{1, 3, 5, 10\}$. The preferred value of r is 5 on all datasets.

2) Impact of m and n: m and n denote the numbers of anomaly and background prompt points, respectively. Their impact on the detection performance is shown in Fig. 16. On the Los Angeles, Gulfport, and Gainesville datasets, too many anomaly prompt points degrade the detection performance. The preferable value of m is 10 on these datasets. In contrast, on the Texas Coast, San Diego, and San Diego-III datasets, a low number of anomaly prompt points results in incomplete anomaly detection. The preferable value of m is 50 on these datasets. In addition, due to the diversity of the background, a larger number of background prompt points tends to improve detection performance on large-scale datasets such as Cri and Segundo. Accordingly, the preferable values of m and n are 10 and 2 on small-scale datasets, while 10 and 200 are recommended on large-scale datasets. Although our experimental datasets cover a wide range of real-world detection scenes, an adaptive parameter selection strategy will be studied in future work.

3) Impact of α: The weight α is assigned to regions outside the candidate boxes generated by Grounding DINO and the target masks. The value of α is set to $\{0.01, 0.05, 0.1, 0.2, 0.5\}$. Experimental results are shown in Table XVI. A value of α that is too small (e.g., 0.01) fails to effectively detect anomalies outside the masks, as observed on the Gainesville dataset. Conversely, a too large value of α (e.g., 0.5) amplifies the interference from complex backgrounds outside the masks, as seen on the San Diego dataset. The preferable value of α is 0.1 on all datasets.

F. Inference Time

Since the foundation models used in our method are frozen, our method does not require training the network. The inference time of all deep learning-based HAD algorithms is

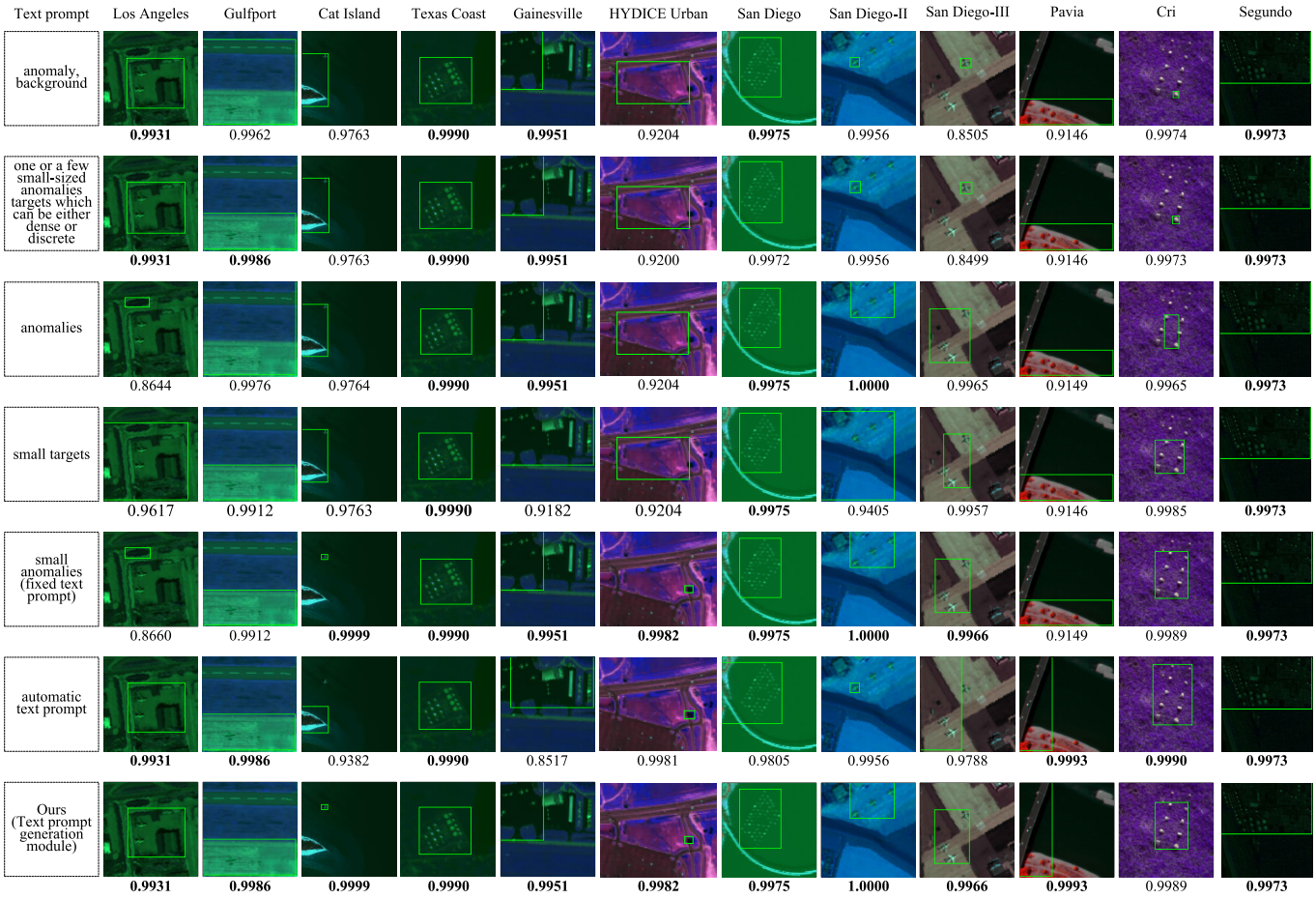

Fig. 14. Effect of different text prompts on all datasets. Green boxes denote target regions delineated by text prompts.

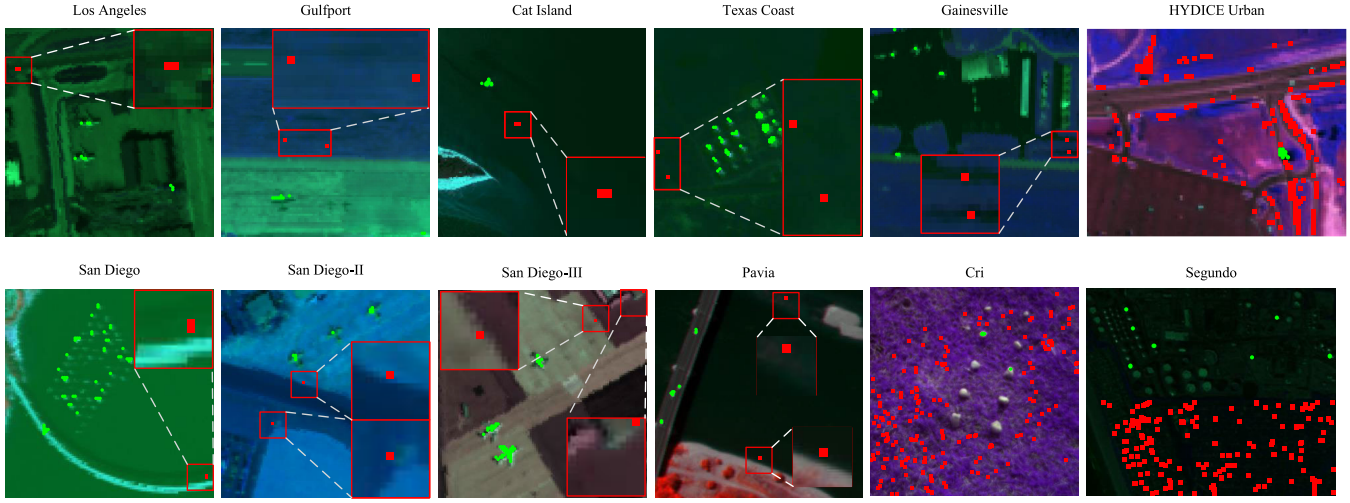

Fig. 15. Visualization of point prompts. The pixels colored in green denote the anomaly point prompts. The pixels colored in red denote the background point prompts, with the red boxes being close-ups.

presented in Table XVII. All methods are performed on a desktop equipped with 32 GB of memory and an NVIDIA RTX 4070Ti graphics processing unit (GPU). It can be observed that our method spends the least time among the foundation model-based methods. A large number of prompt points in HyperFree increase the inference time of this model. Compared with other methods, our method achieves a better trade-off between efficiency and accuracy.

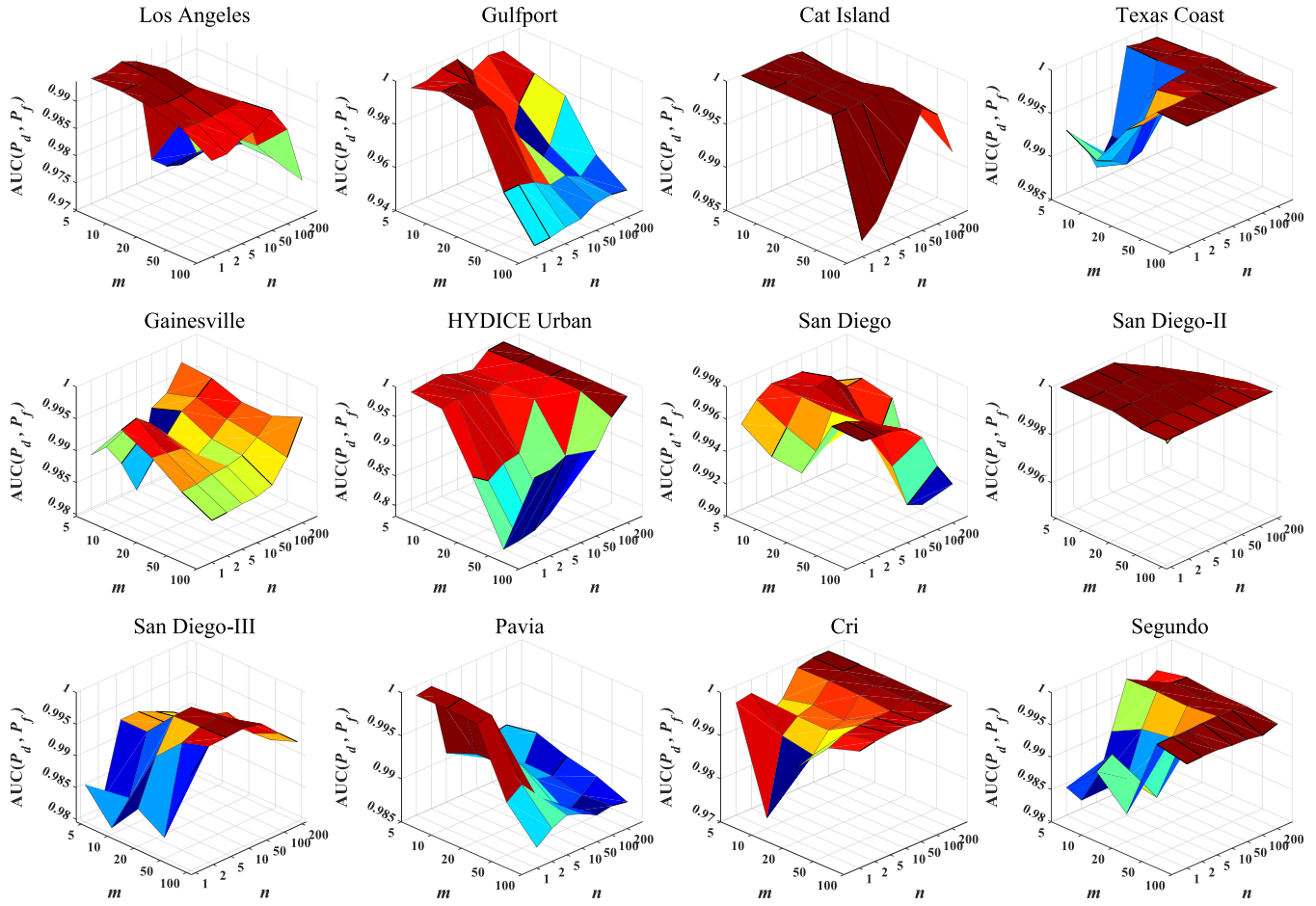

Fig. 16. Impact of m and n on our method.

TABLE XVII
INFERENCE TIME (SECOND) OF DIFFERENT DEEP LEARNING METHODS

Method	Los Angeles	Gulfport	Cat Island	Texas Coast	Gainesville	HYDICE Urban	San Diego	San Diego-II	San Diego-III	Pavia	Cri	Segundo
AutoAD	0.011	0.011	0.014	0.012	0.012	0.011	0.012	0.012	0.012	0.011	0.044	0.033
BockNet	0.014	0.013	0.028	0.016	0.012	0.029	0.012	0.016	0.011	0.098	0.094	0.107
GTHAD	0.061	0.031	0.064	0.031	0.035	0.049	0.030	0.031	0.030	0.093	0.486	0.346
MSNet	0.002	0.003	0.004	0.002	0.002	0.002	0.002	0.002	0.002	0.003	0.030	0.009
EGSSFD	18.866	18.018	27.252	20.568	19.590	18.396	17.114	12.621	18.915	22.283	281.306	135.339
OTAD	0.022	0.014	0.041	0.015	0.014	0.015	0.015	0.014	0.014	0.025	0.181	0.146
SGLNet	2.148	1.922	4.261	1.974	2.081	1.523	2.085	2.052	2.057	4.182	1.054	14.925
HyperSIGMA	0.313	0.304	0.691	0.310	0.310	0.239	0.309	0.318	0.313	0.682	4.250	1.869
HyperFree	27.430	33.181	29.064	10.207	12.158	27.041	10.345	24.027	27.296	29.882	12.324	13.897
Ours	0.694	0.598	0.677	0.623	0.629	0.709	0.643	0.616	0.619	0.718	0.737	0.667

V. CONCLUSION

In this article, we propose a dual prompt-guided HAD method for cross-scene generalization by viewing HAD as a segmentation task. A clustering-based band selection method is proposed to align the HSI channels with SAM. The text prompt generates masks by Grounding DINO to indicate the potential anomaly and background regions. The point prompt strategy selects the pseudo anomaly and background points according to the masks and guides SAM to perceive the anomaly target and background. Experimental results demonstrate that our method achieves superior cross-scene generalization performance to existing state-of-the-art HAD methods without additional training.

Our method achieves satisfactory cross-scene detection performance. However, since it uses large-parameter foundation models as the backbone, its inference time is inevitably longer than that of lightweight deep learning methods, which may hinder its application to real-time scenes. Besides, the text

prompt generation module possesses heuristic properties. Our ensemble strategy can alleviate the domain shift problem of the text prompt generation module, but it cannot completely eliminate the hallucination caused by domain shift, which is an inherent limitation of current foundation models. Although developing foundation models dedicated to hyperspectral data may mitigate the domain shift problem, as stated in the Introduction, our work focuses on addressing the generalization issue of HAD under resource constrained conditions. In the future, we will study cross-modal knowledge distillation to obtain a lightweight network for faster inference.

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